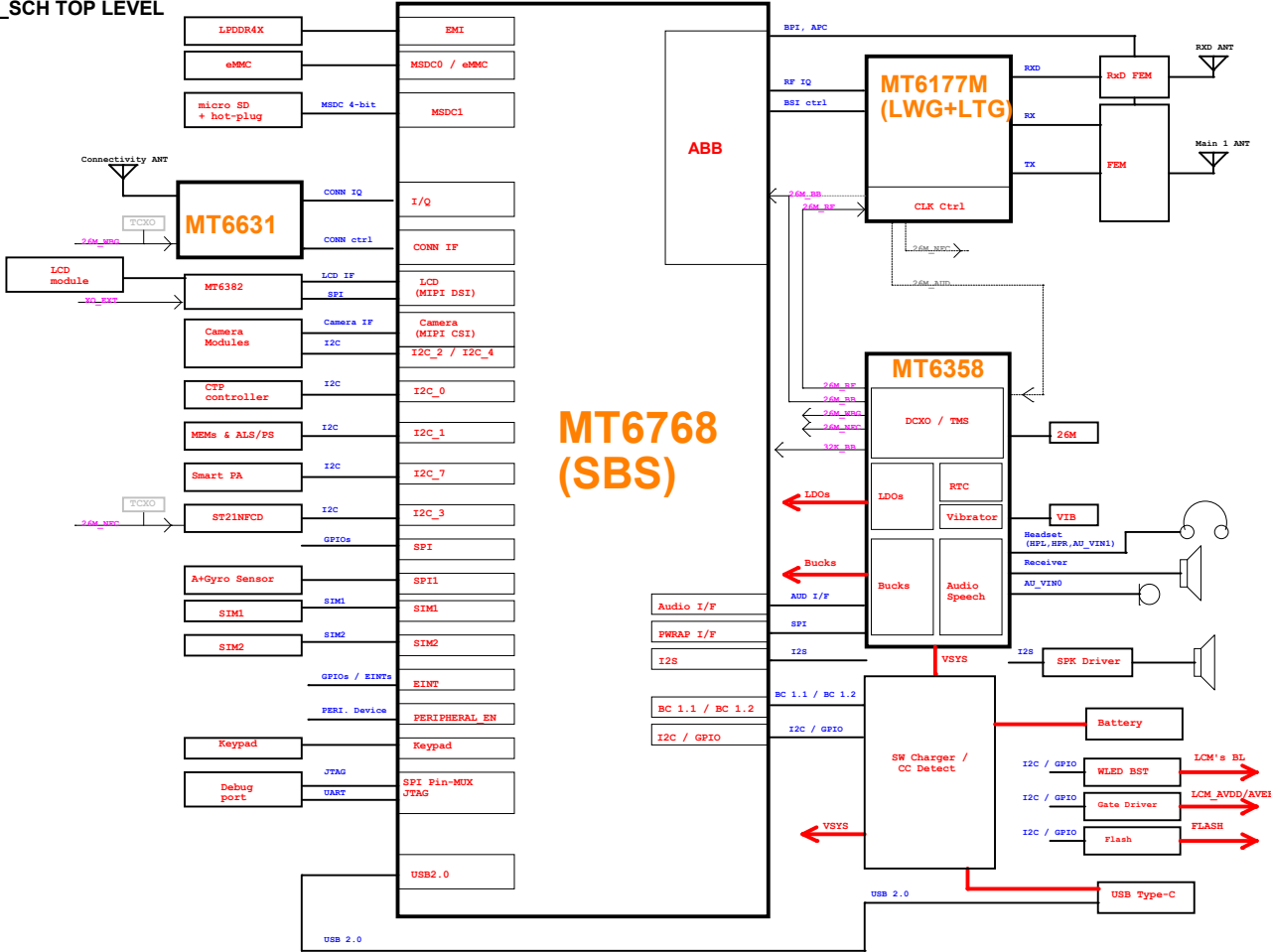
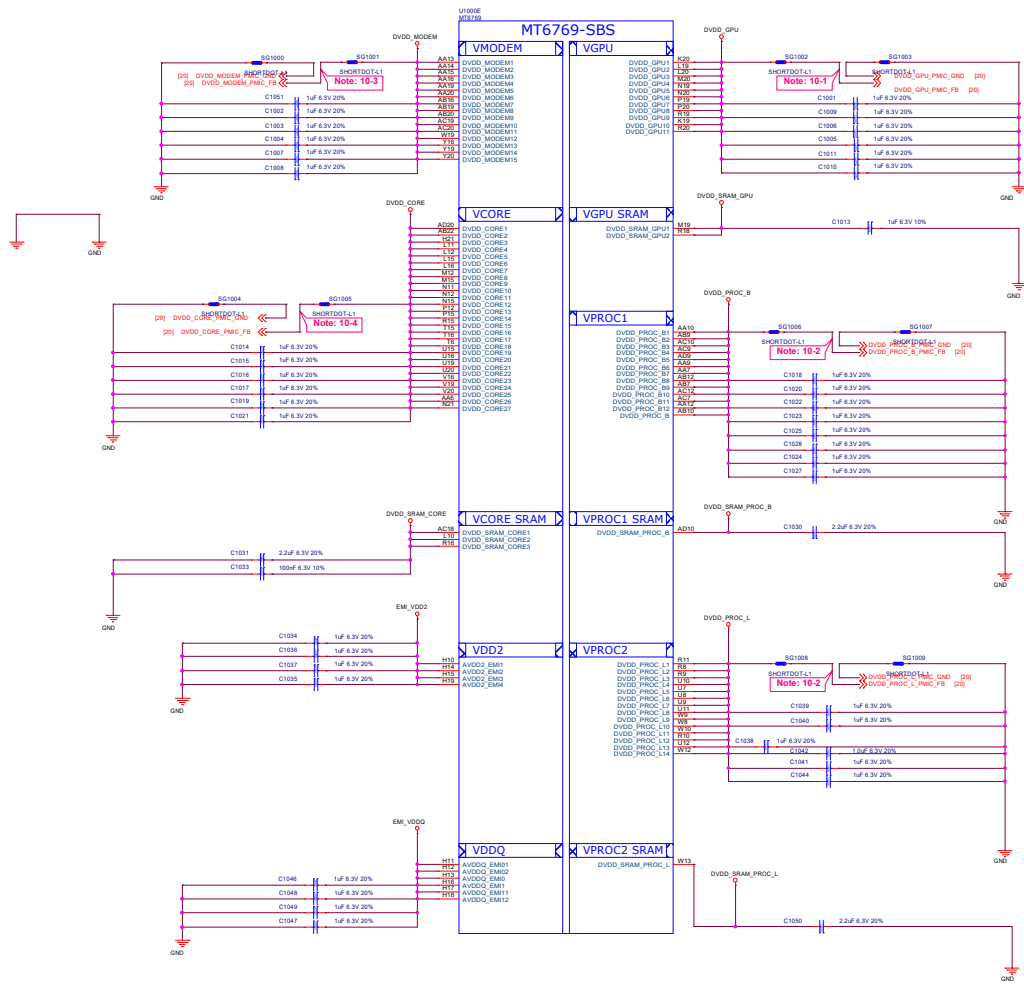




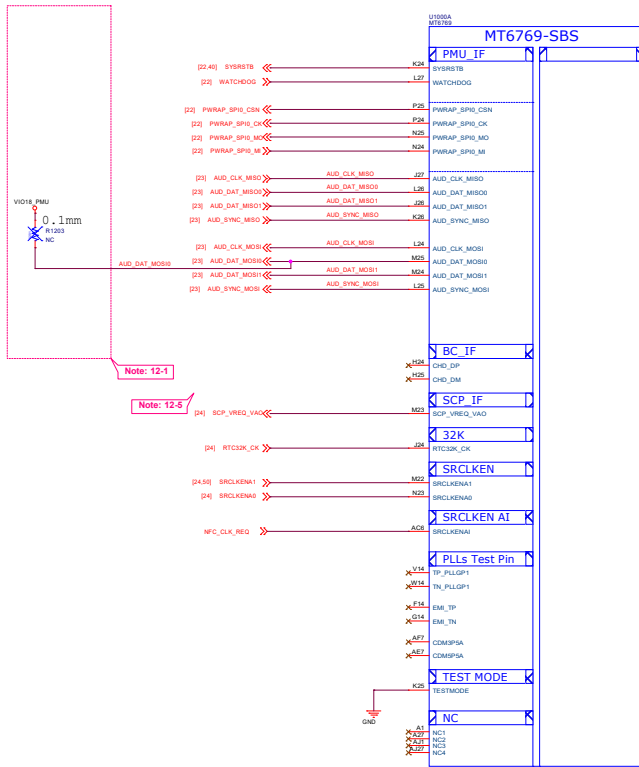
LongCheer		
Title 01_Block_Diagram		
Size D	Project M19_MB_V1	Rev V1
Date: Friday, March 08, 2024		Sheet 1 of 78

Schematic design notice of "10_BB_POWER_PDN" page:



- Note 10-1: Differential pair of DVDD_GPU remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-2: Differential pair of DVDD_PROC remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-3: Differential pair of DVDD_MODEM remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)
- Note 10-4: Differential pair of DVDD_CORE remote sense must be close to BB's ball.
Remote sense trace with GND shielding to PMIC (Differential)

LongCheer		
Title BB_POWER_PDN		
Size D	Project C3N	Rev v1
Date: Tuesday, April 09, 2024		Sheet 10 of 78



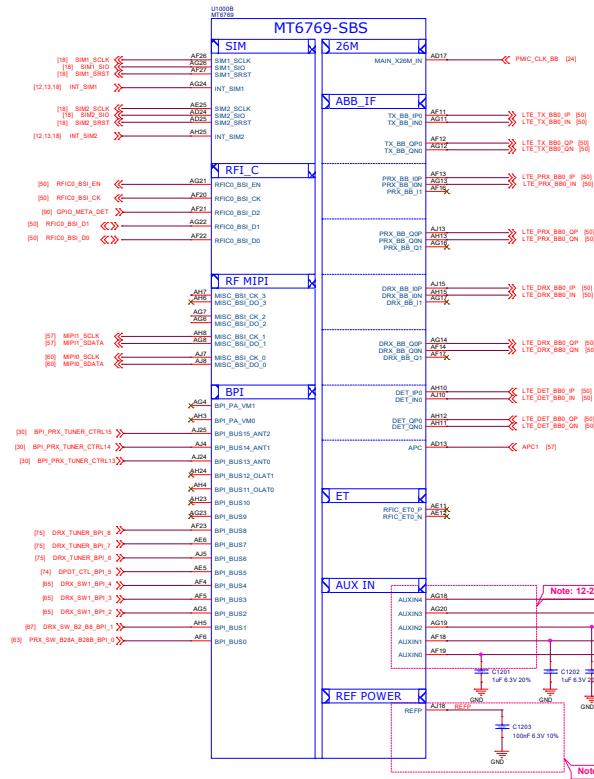
Schematic design notice of "12_BB_1" page:

Note 12-1: PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pins to select which interface will be the JTAG pin out

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	JTAG Function
default=PU	default=PD	AP_JTAG
HI	LO	N/A
HI	HI	SPI0 + EINT8
LO	LO	SPI0 + EINT8
LO	HI	N/A

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: The de-coupling cap. for REFP (AJ18 ball) have to be placed as close to BB as possible.

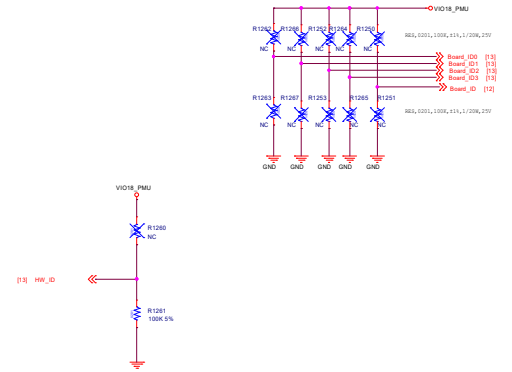


Note 12-4: HW pin for DDR type feature in bootstrap (refer to BB HW design Notice)

AUD_SYNC_MISO	AUD_CLK_MISO	CAM_PDN3	PMIC 6358 voltage	DDR Type
default=PD	default=PD	default=PD	VDRAM1 / VDRAM2	DDR
No ext. pull	No ext. pull	No ext. pull	1.125v / 0.6v	LP4X eMCP
No ext. pull	12K pull to VIO18	No ext. pull	OFF / 1.8v	Reserved
12K pull to VIO18	No ext. pull	No ext. pull	1.225v / OFF	LP3 eMCP
12K pull to VIO18	12K pull to VIO18	No ext. pull	1.125v / 1.8v	Reserved

Note 12-5: Charger must have D+/D- pin for charger type USB detection.
Charger must have at least 500mA USB current for All charger type.

1. R1208 must close to L1E Band 7 PA or the bottom PA ~2mm.
2. The distance is the shortest distance from package edge to edge.



Thermistor to sense AP temperature

1. R1207 must keep a distance about 6-8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

LongCheer

Title 12_BB_1

Size D

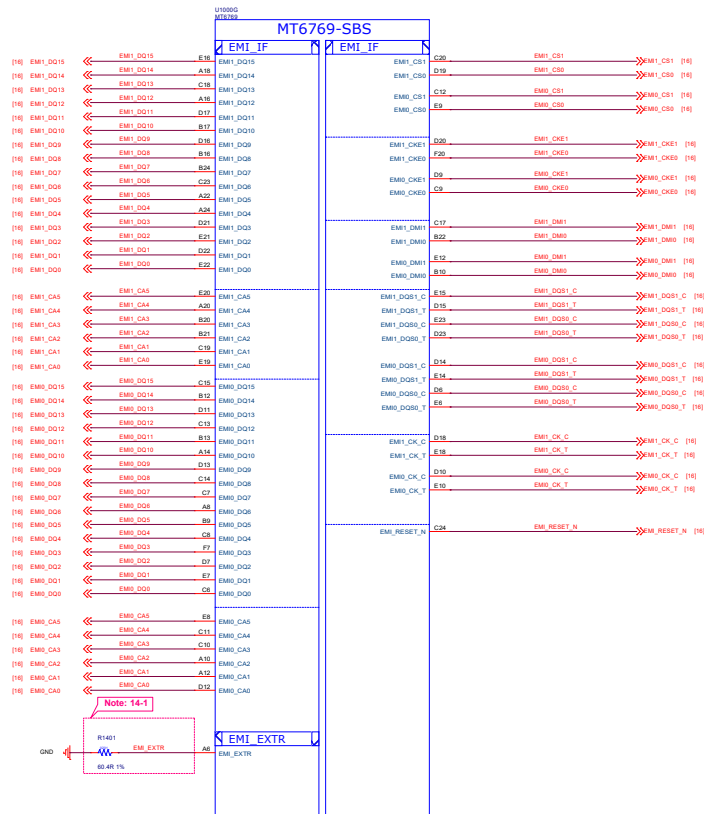
Project

M19_MB_V1

Rev V1

Date: Tuesday, April 09, 2024

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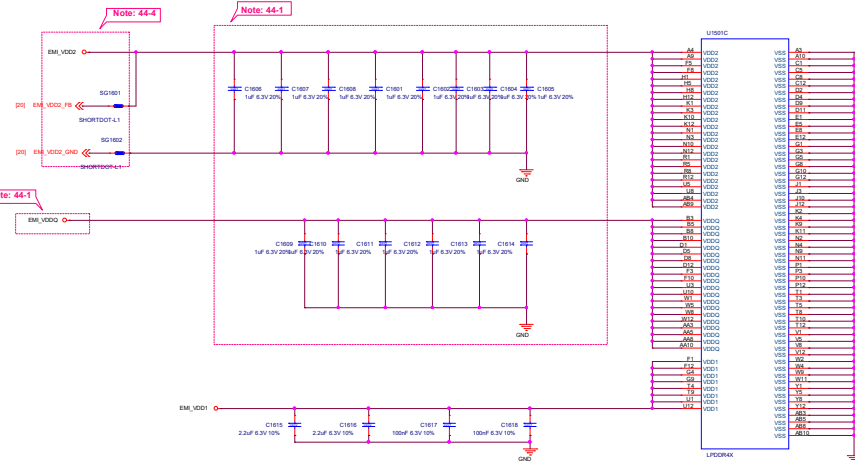
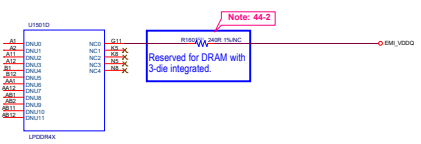
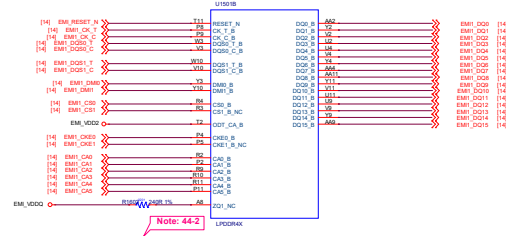
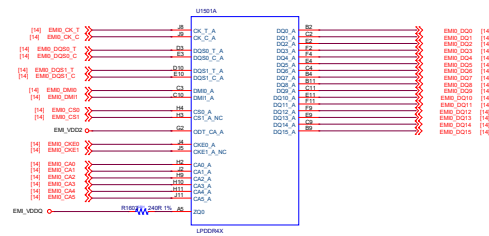


Schematic design notice of "14_BB_3" page.

Note 14-1: R1401 please select 60.4 ohm (1%) resistor

LongCheer			
Title		14_BB_3	
Size	Project	M19_MB_V1	Rev
D			V1
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LongCheer		
Title 14_BB_3		
Size A2	Project M19_MB_V1	Rev V1
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Schematic design notice of "44_Memory_eMMC_LPDDR4X"

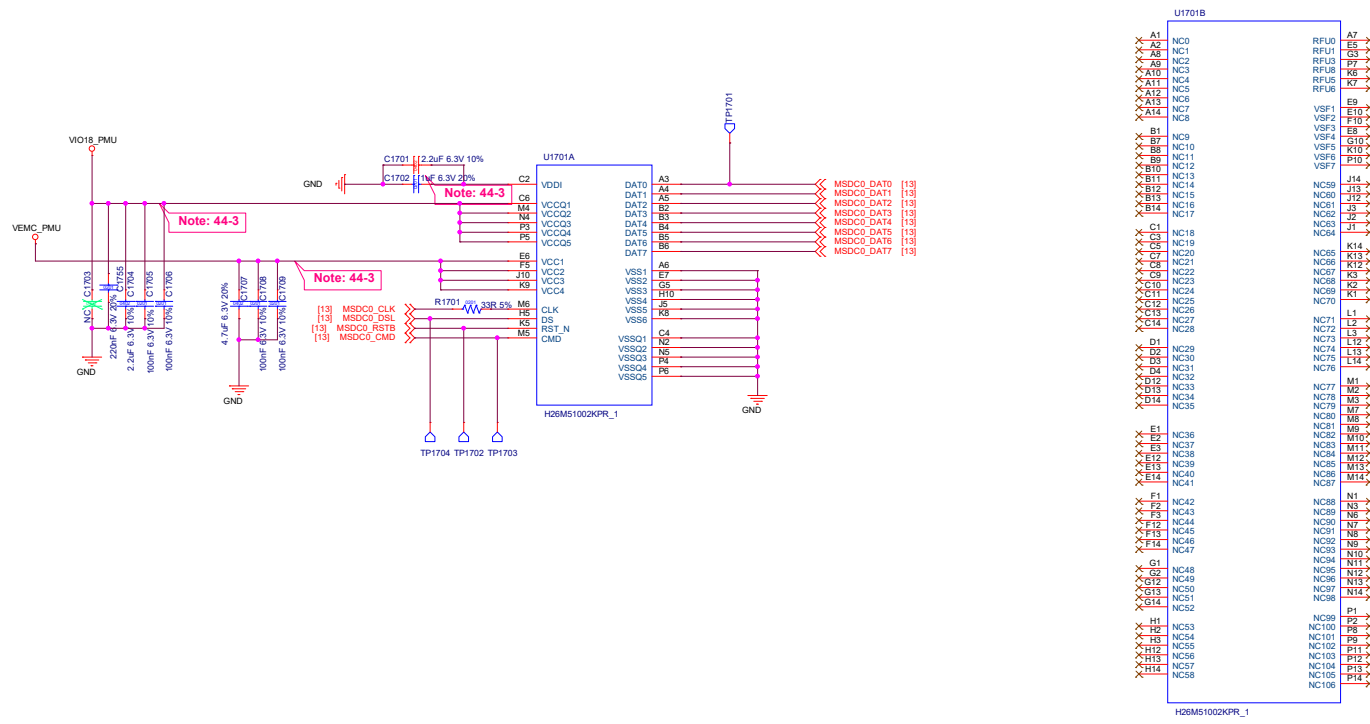
Note 44-1: Please refer to power supply related page select VDRAM 2 / VDRAM1 output voltage properly for LPDDR4X

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ.

Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

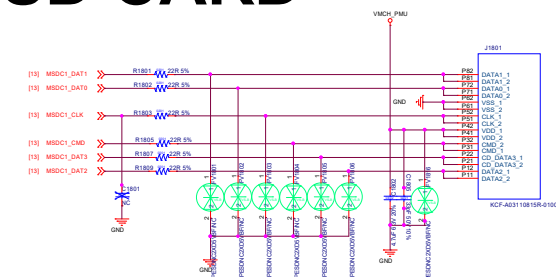
Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC >10uF, at PMIC page];
please also refer to MMD and layout guide for placement.

LongCheer			
Title Memory_LPDDR4X			
Size	Project		Rev
A1	M19_MB_V1		V1
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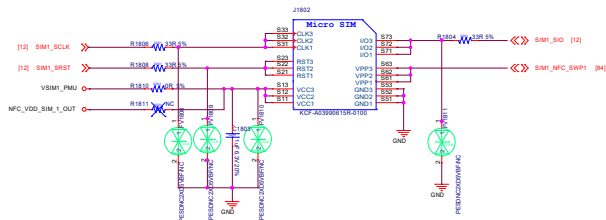


LongCheer			
Title Memory_eMMC			
Size	Project	M19_MB_V1	
A2			Rev V1
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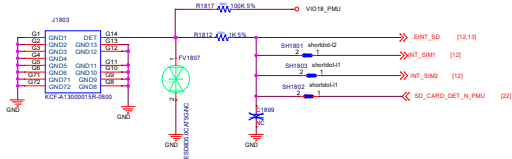
SD CARD



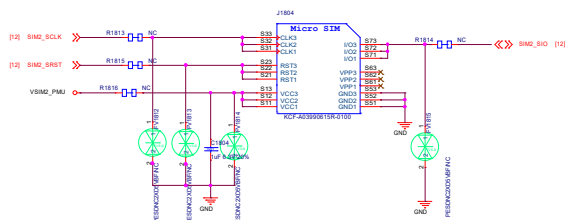
SIM1 PCB靠板边



Detect



SIM2 PCB靠板内

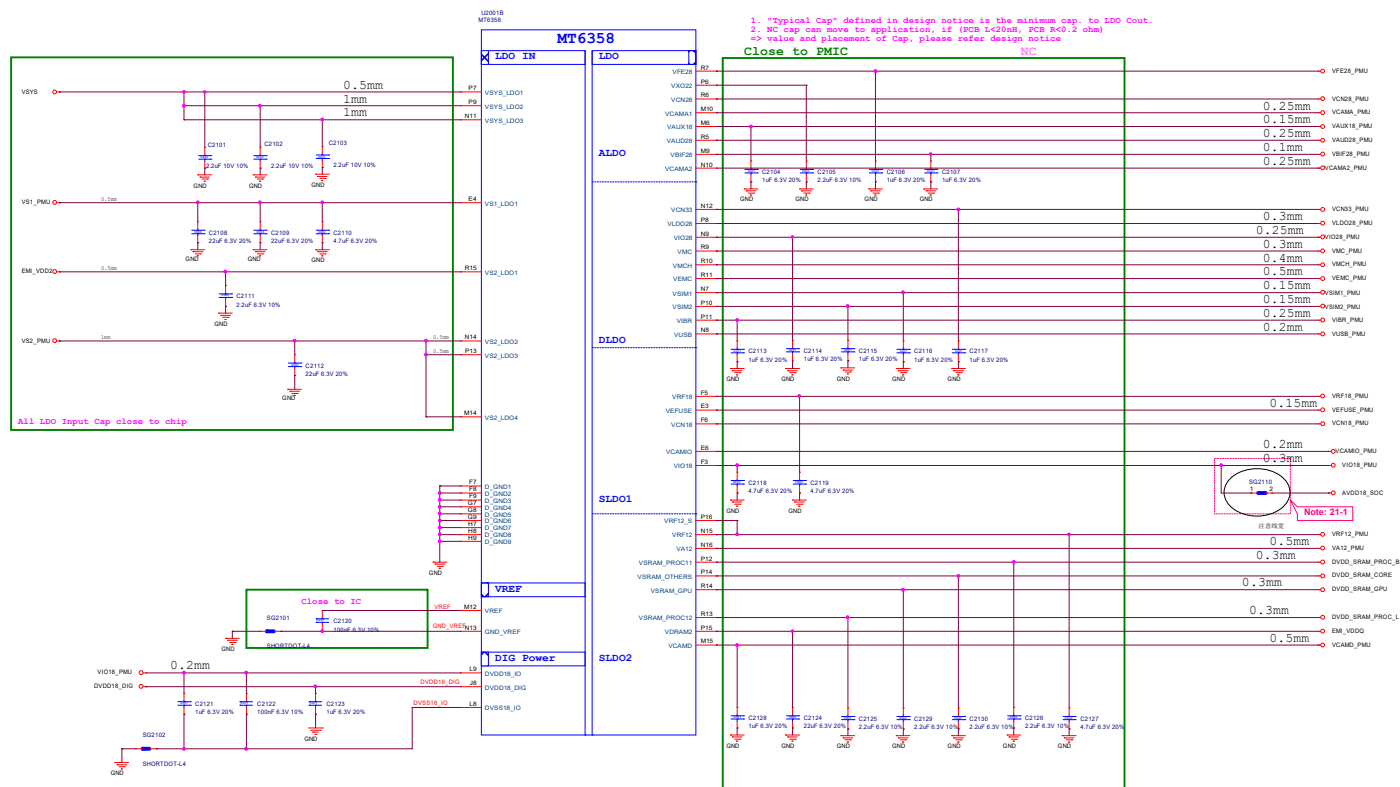


Note 46-1:

Schematic design notice of "46_MEMORY_SD Card" page.

For better ESD performance, please select suitable components for system protection.

LongCheer		
Title Memory_SD_SIM_Card		
Size D	Project M19_MB_V1	Rev V1
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Schematic design notice of "21_POWER_MT6358-LDO" page.

Note 21-1: Please set SG2110 close to C2118, making star connection between VIO18_PMU and AVDD18_SOC near to LDO cap. C2118
Please also refer to MT6358 design notice for further detail design information

Note 21-2: If these power trace can meet LDO layout constraint, these CAP can be NC or removed.
Please refer to MT6358 design notice.

Note: 21-2

LongCheer

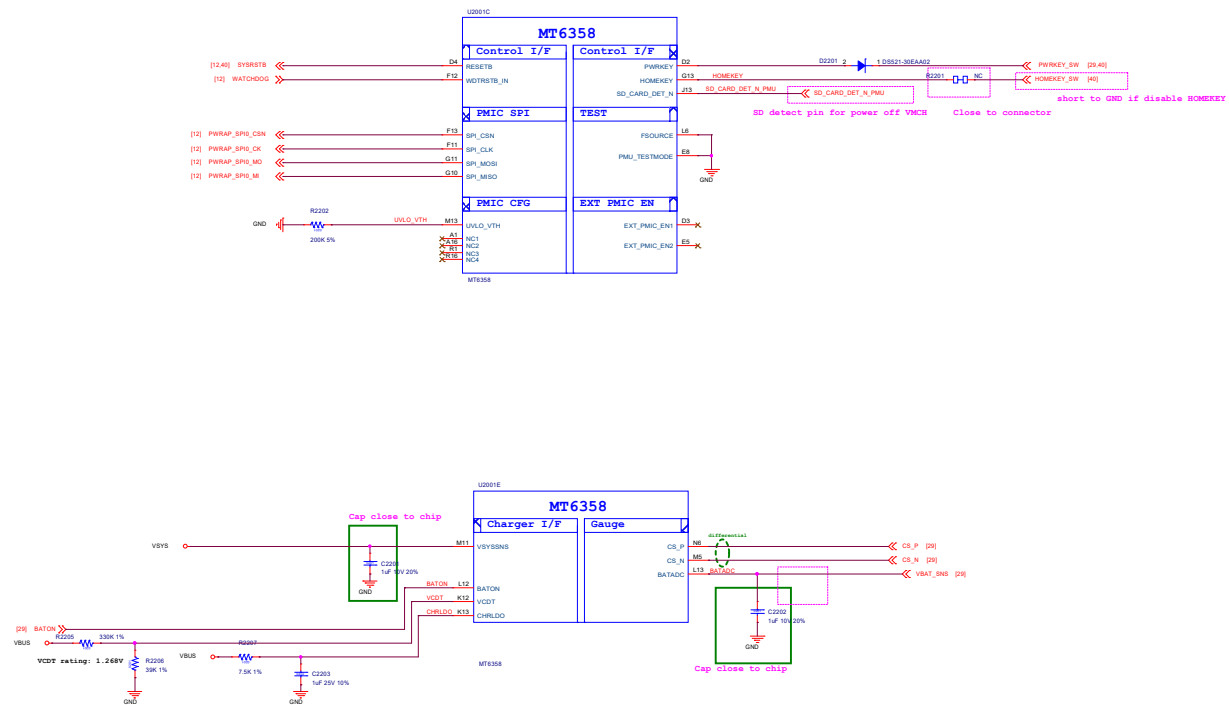
Title POWER_MT6358-LDO

Size D Project M19_MB_V1

Rev V1

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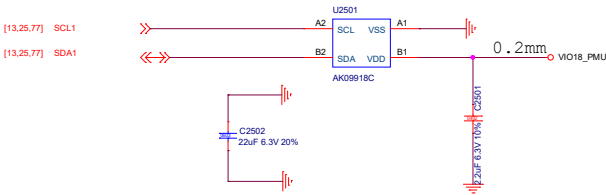


LongCheer		
Title POWER_MT6358-IF		
Size D	Project M19_MB_V1	Rev V1
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LongCheer			
Title POWER_MT6358_Clock			
Size D	Project M19_MB_V1		Rev v1
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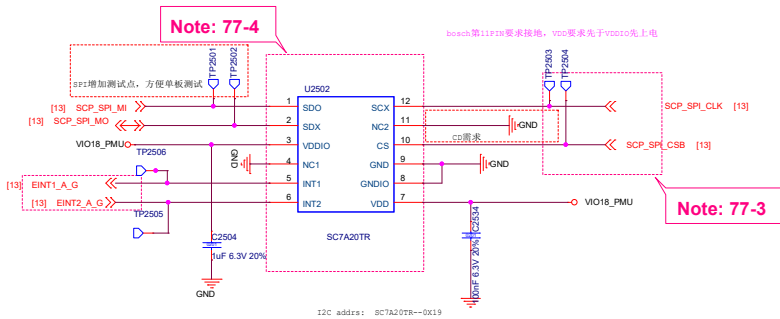
M-Sensor



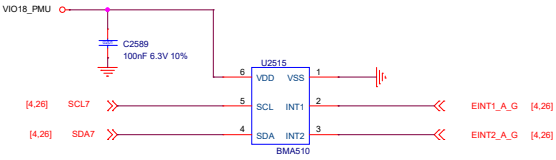
ALS P/L AK09918C

厂家	I2C ADDRESS
AK09918C	0ch

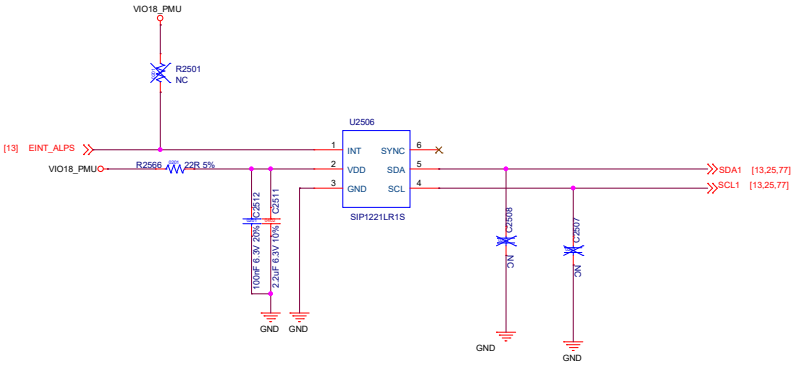
A-Sensor



A sensor BMA510

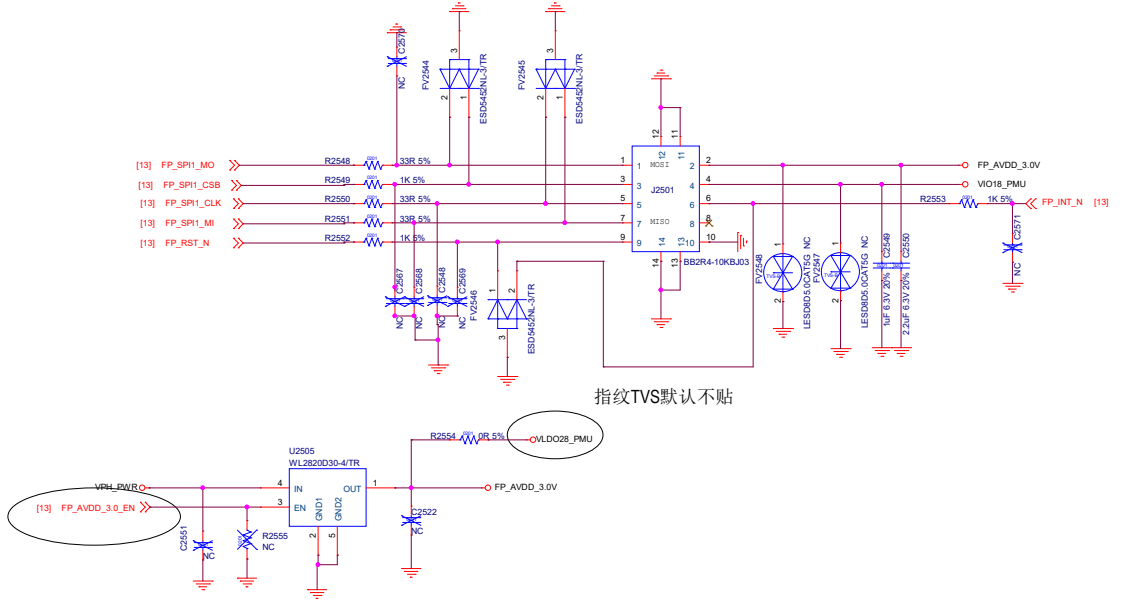


L-SENSOR

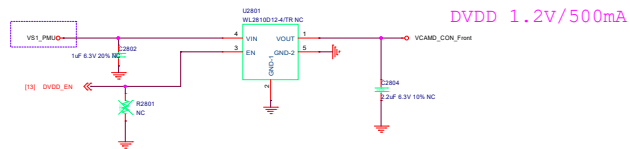


厂家	I2C ADDRESS
TSL25223	0x39

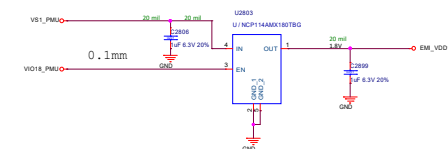
FP-Sensor



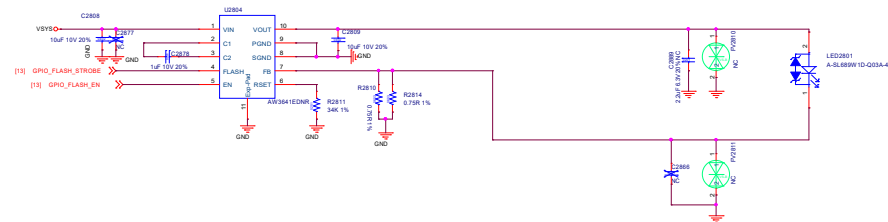
LDO for Front Camera (VCAMD_EXT1) 1.2V



Ext. EMI_VDD1 for LPDDR4X VDD1



FLASH LED Driver



LongCheer

Title POWERS_CAM_FL_DDR

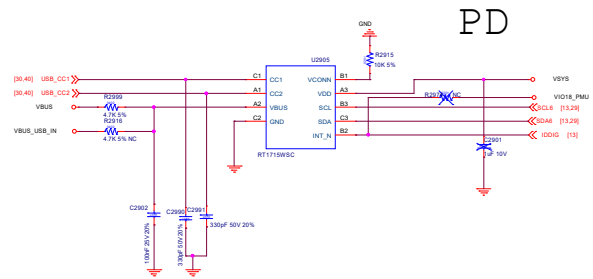
Size D Project M19_MB_V1

Rev v1

Date: Tuesday, March 19, 2024

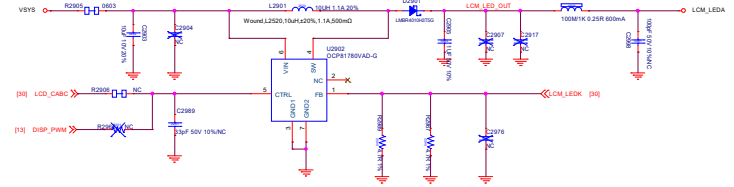
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厂牌	12C 6038030 1 BAT
C098511	0342
MTS617030	0322

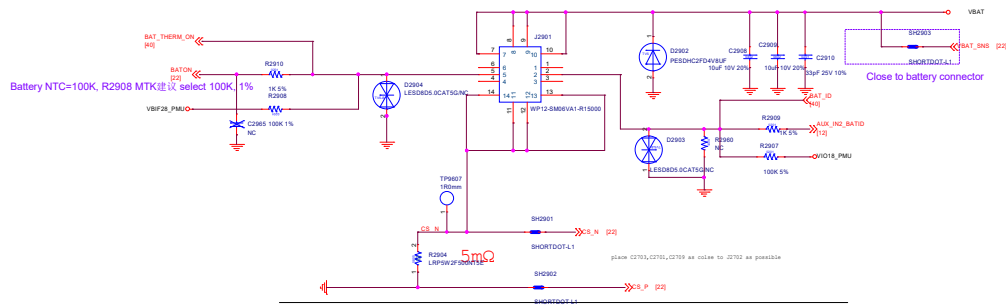


PD

LCM_Backlight



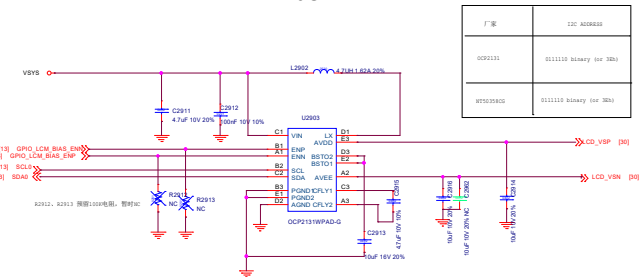
BATTERY CONNECTOR



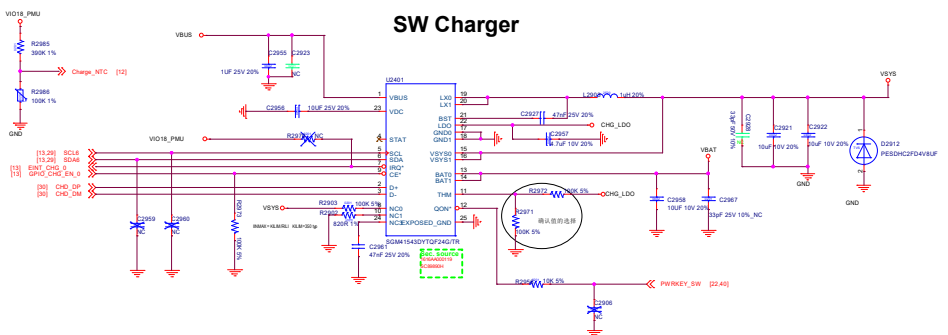
Schematic design notice of "29_EXT_WLED+DB+FL+CHG+PD" page.

Note 29-1: For better ESD or surge performance we need choose suitable device for system protection. Please refer to [Surge device selection guide V2.0] provide by MTK.

LCM Bias



SW Charger



LongCheer

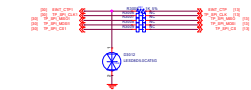
Title EXT_BL+BIAS+CC+CHG+BAT

Size D Project M19_MB_V1

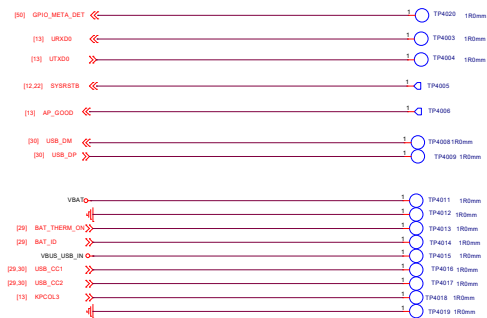
Rev V1

Date: Friday, April 19, 2024

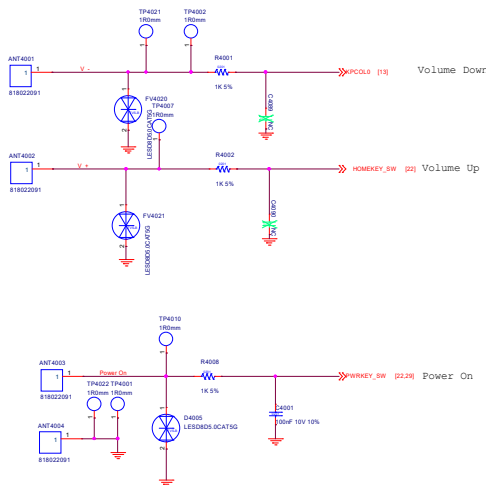
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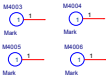
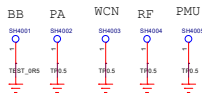
测试点



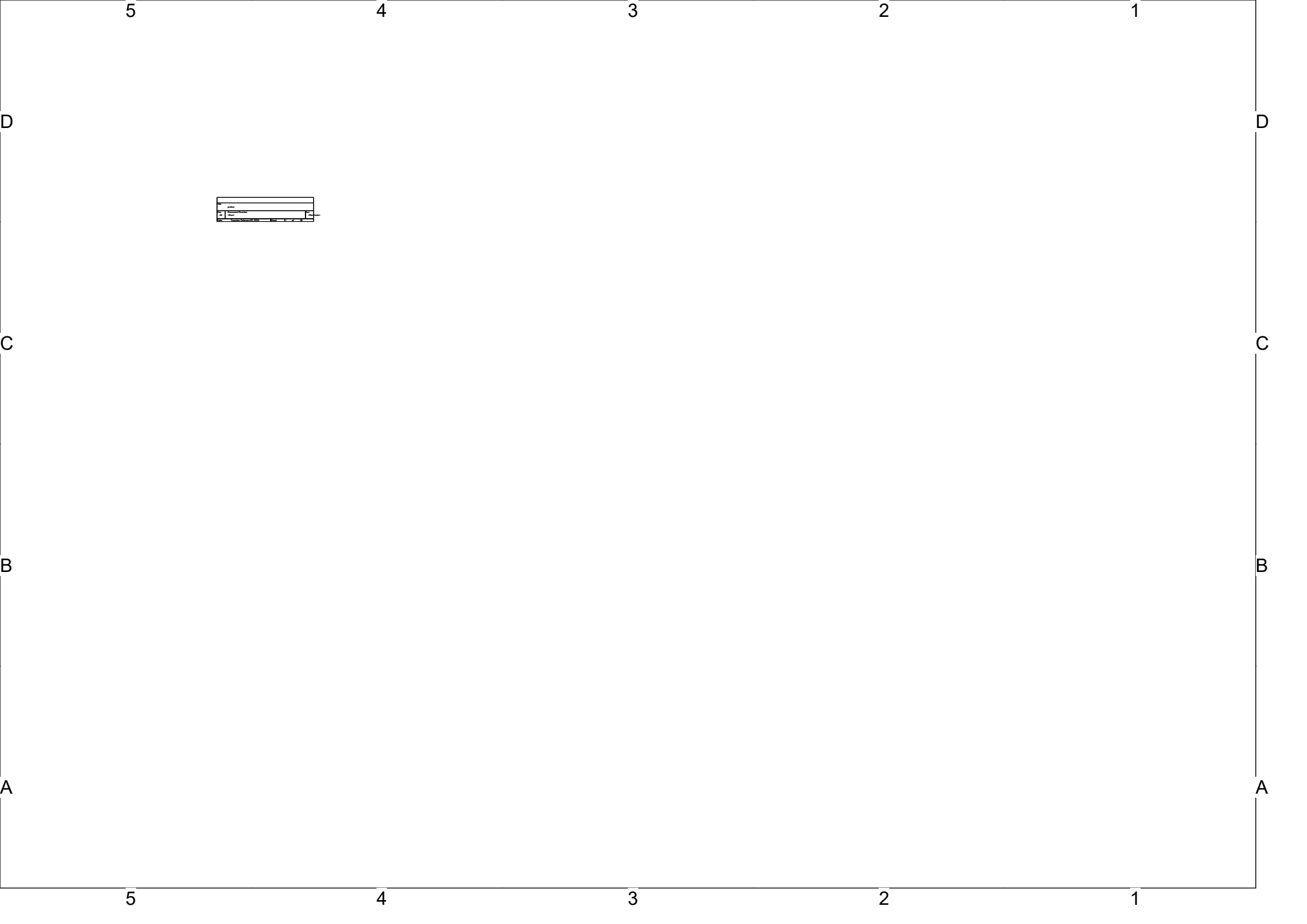
KEY BTB

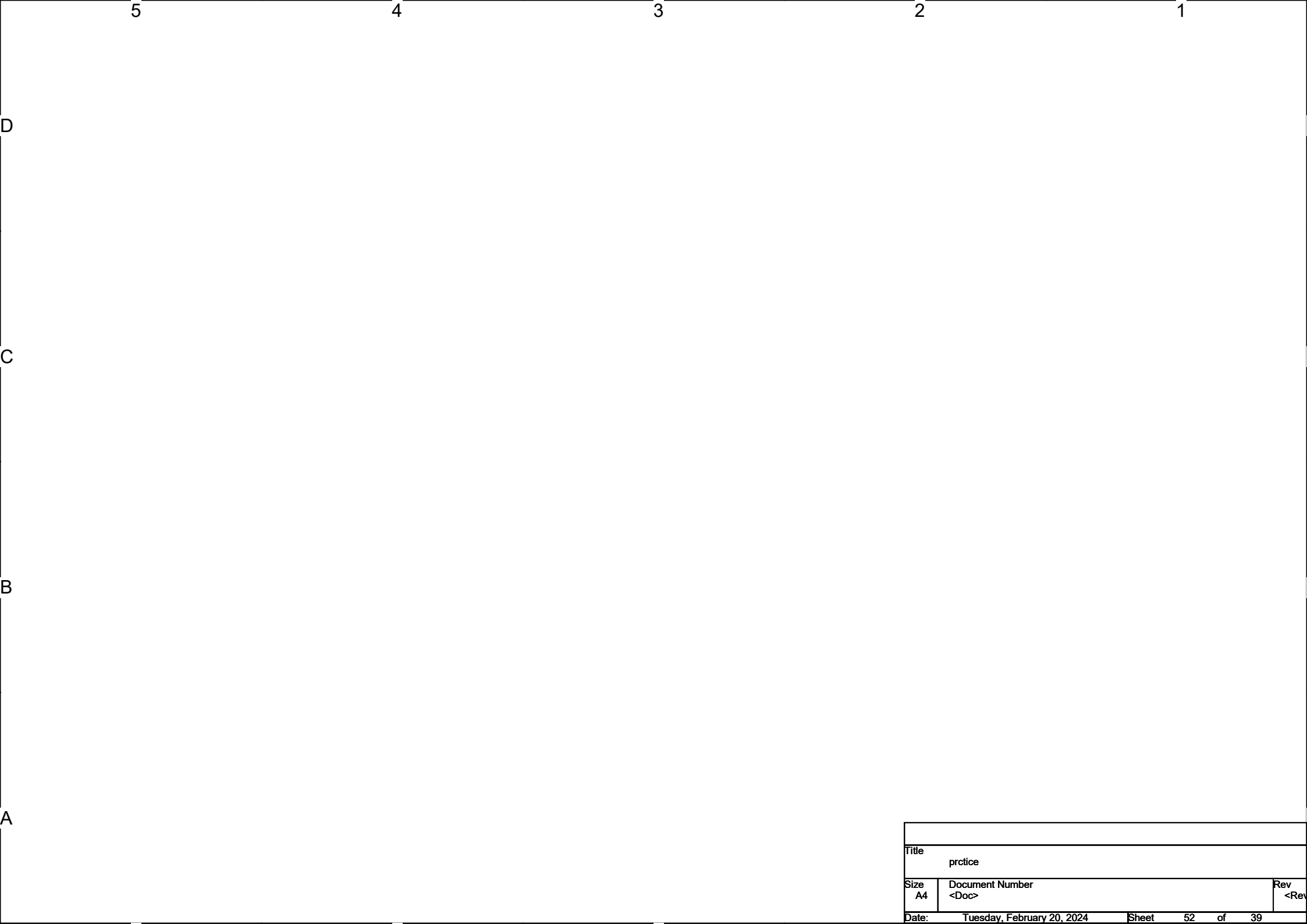


屏蔽罩

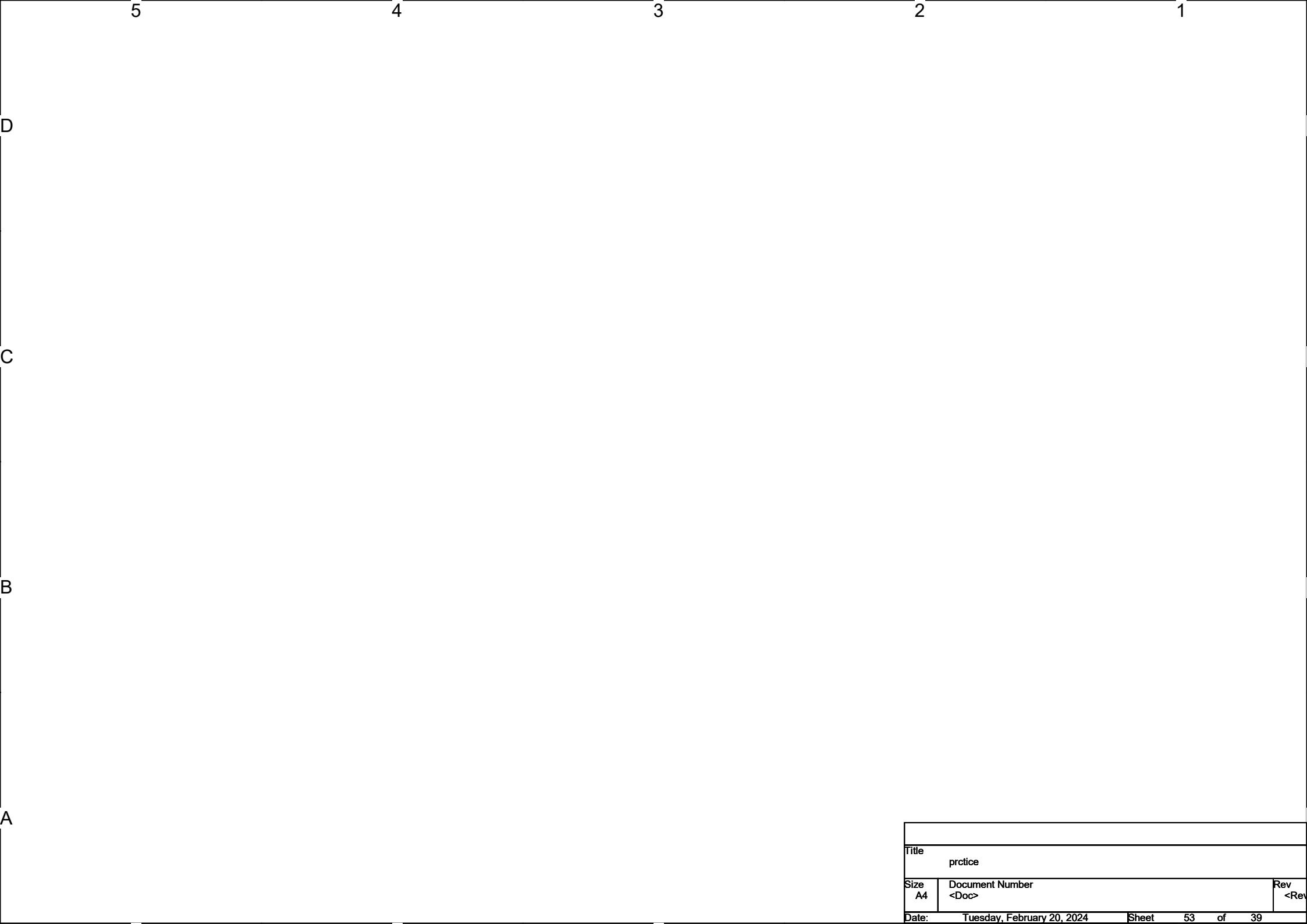


LongCheer		
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Size D	Project M19_MB_V1	Rev v1
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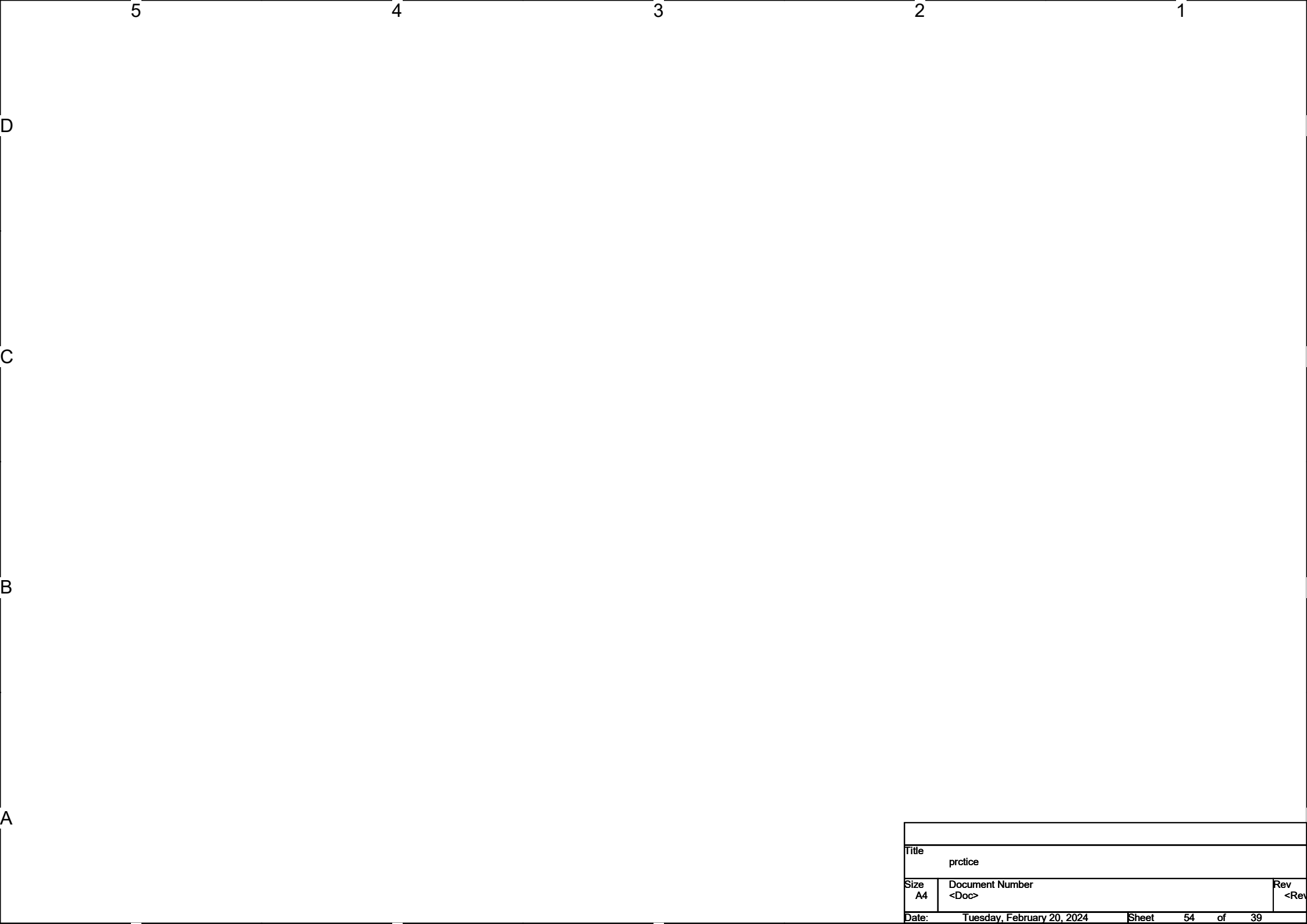




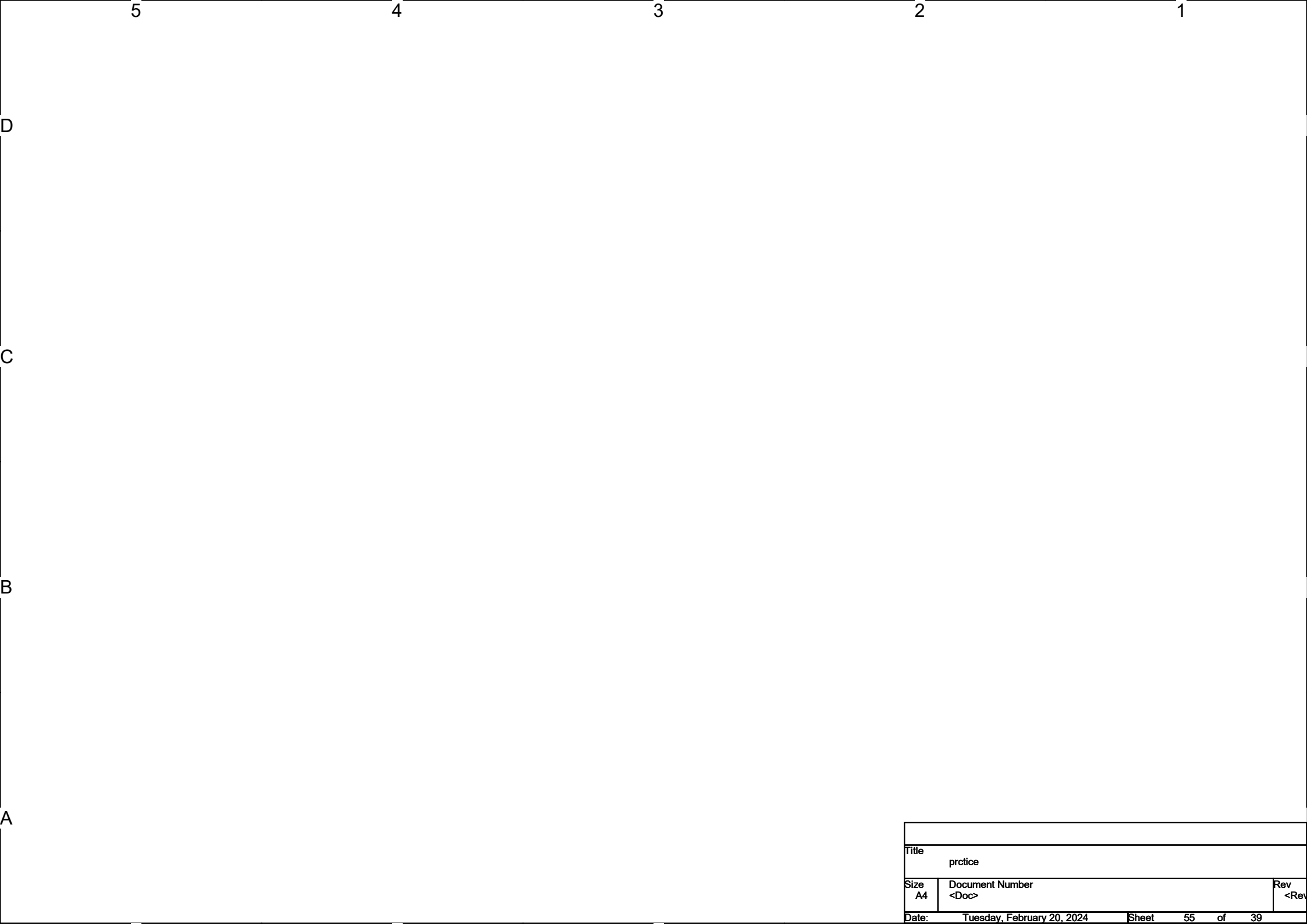
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Size	Document Number	Rev
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Date:	Tuesday, February 20, 2024	Sheet 52 of 39



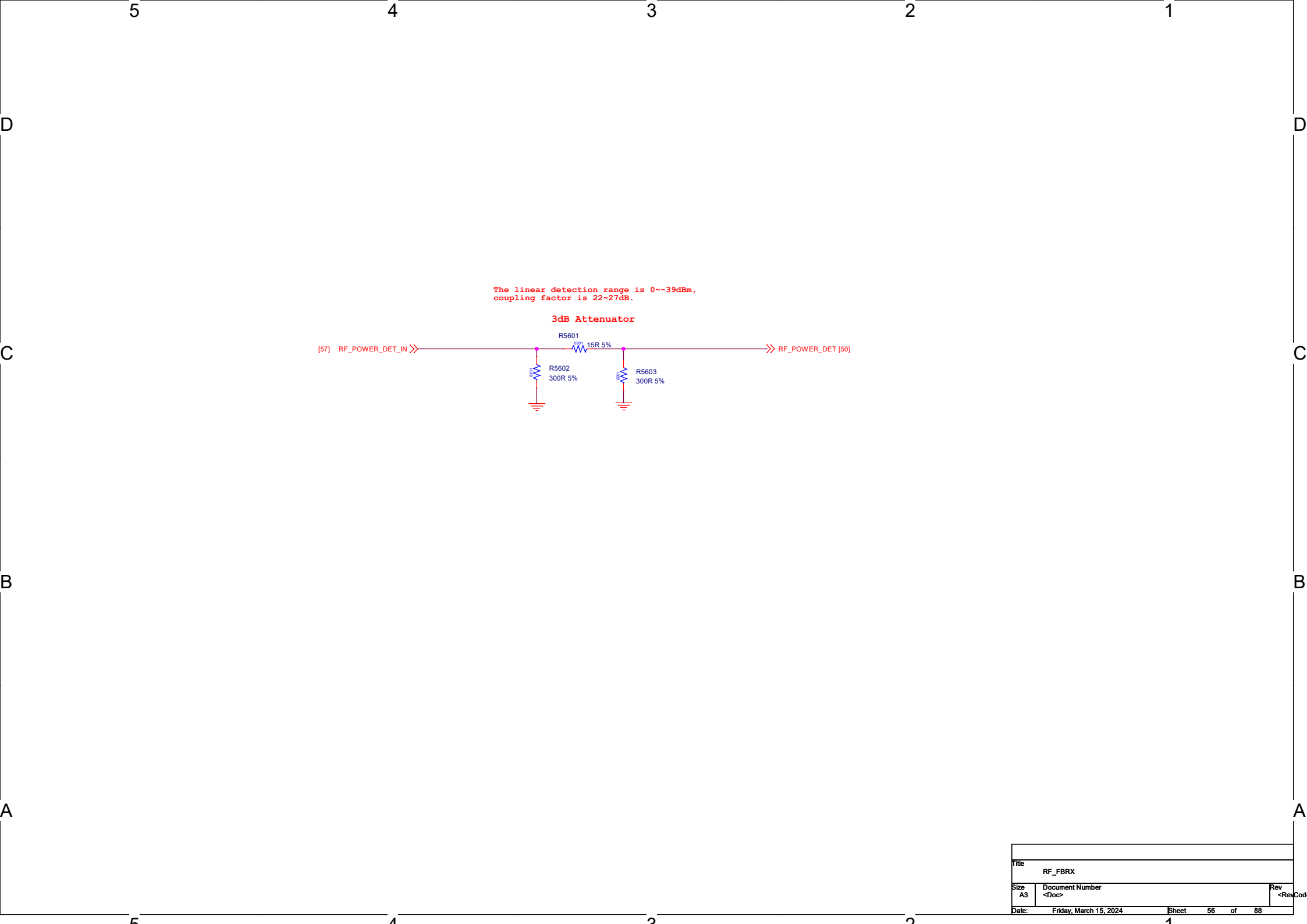
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Date:	Tuesday, February 20, 2024	Sheet 53 of 39

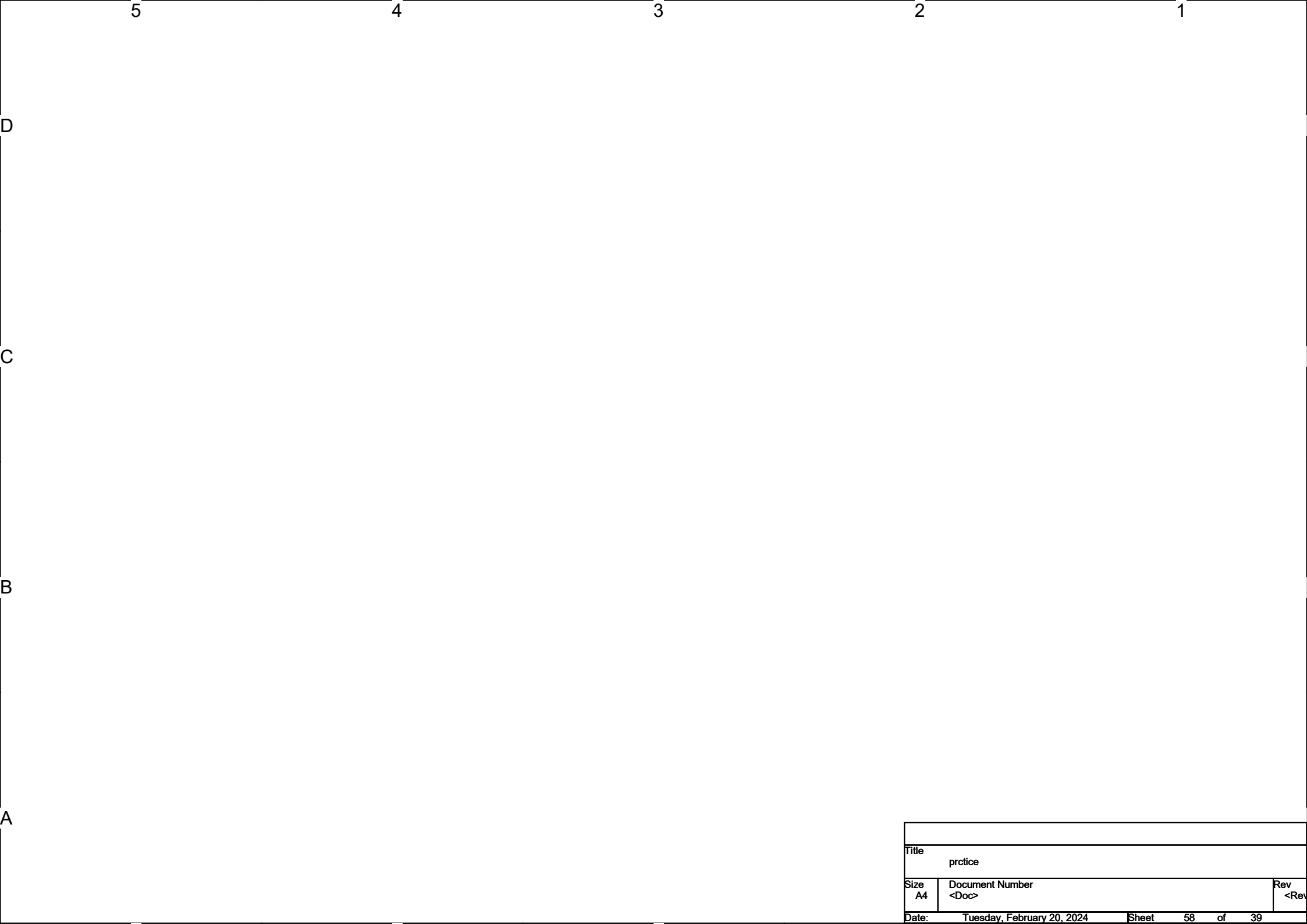


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Date:	Tuesday, February 20, 2024	Sheet 54 of 39

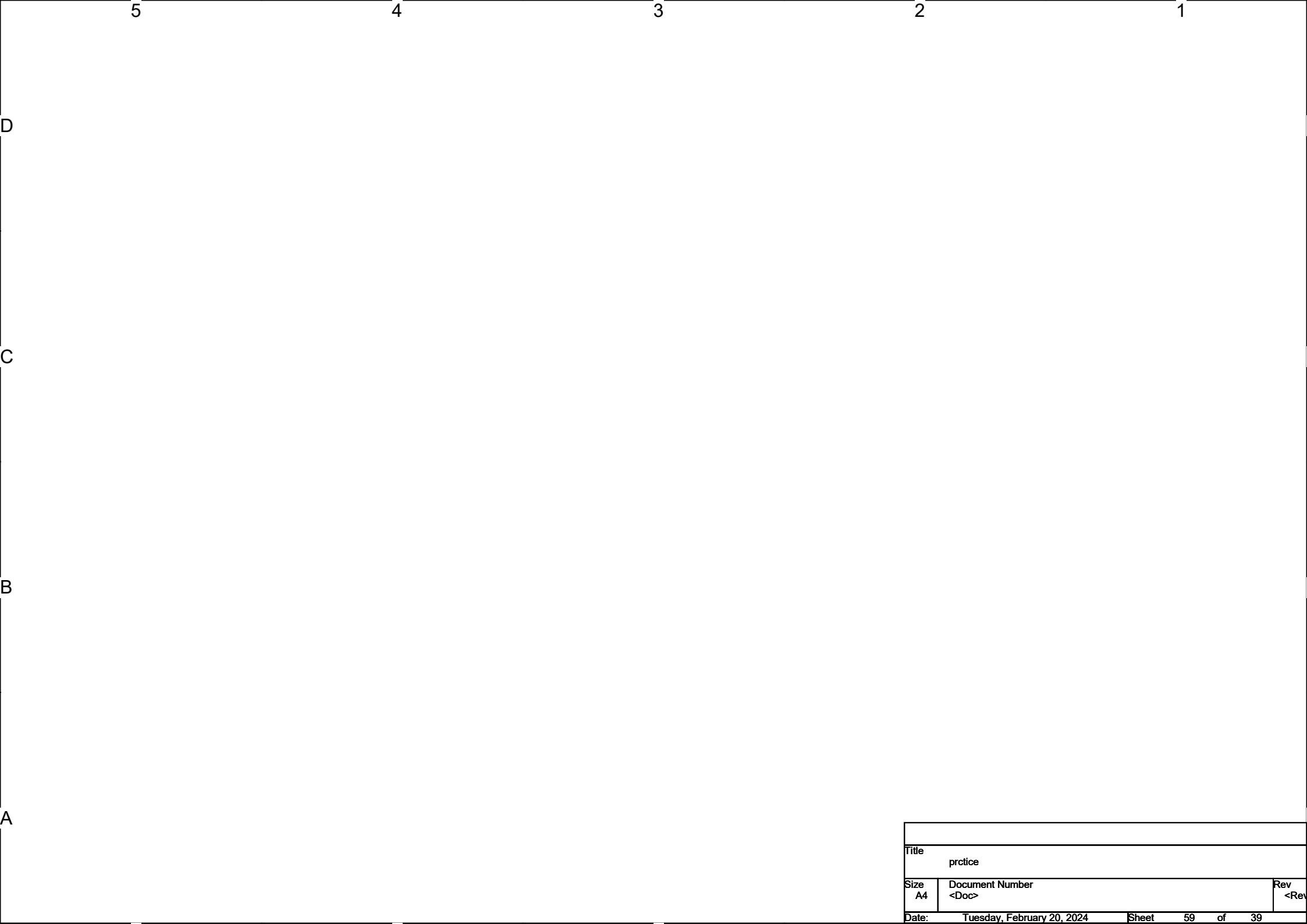


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Date:	Tuesday, February 20, 2024	Sheet 55 of 39

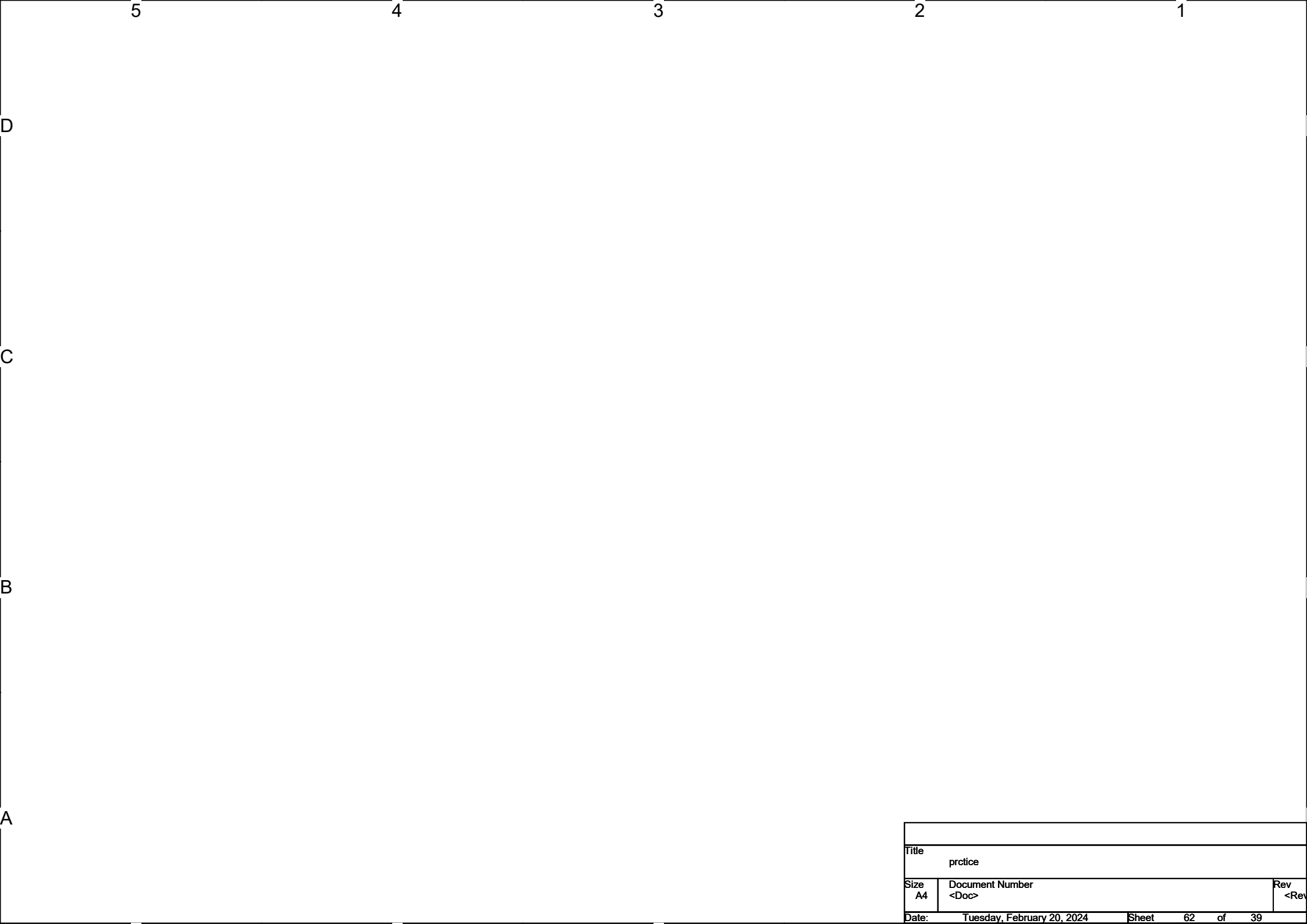




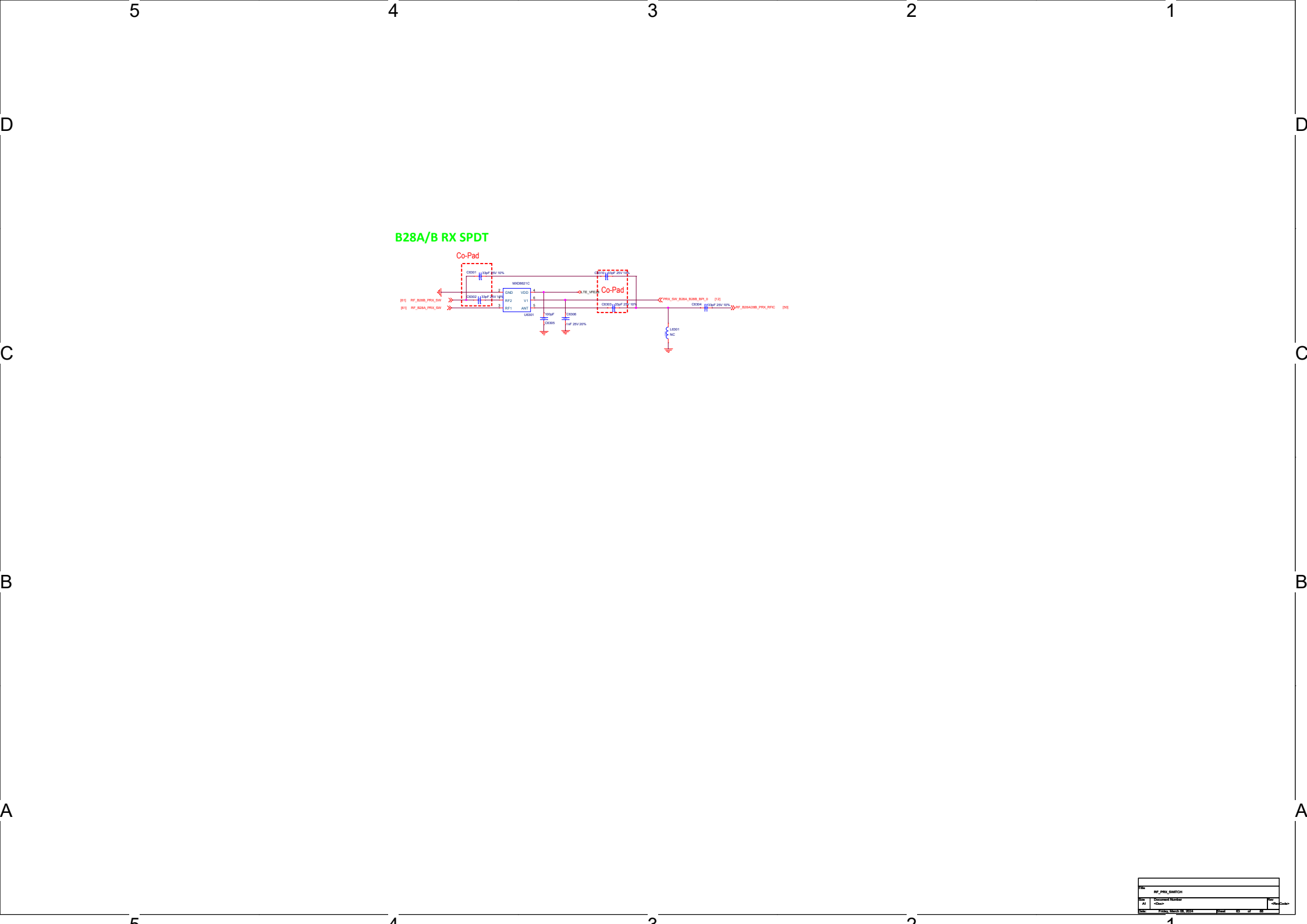
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Date:	Tuesday, February 20, 2024	Sheet 58 of 39

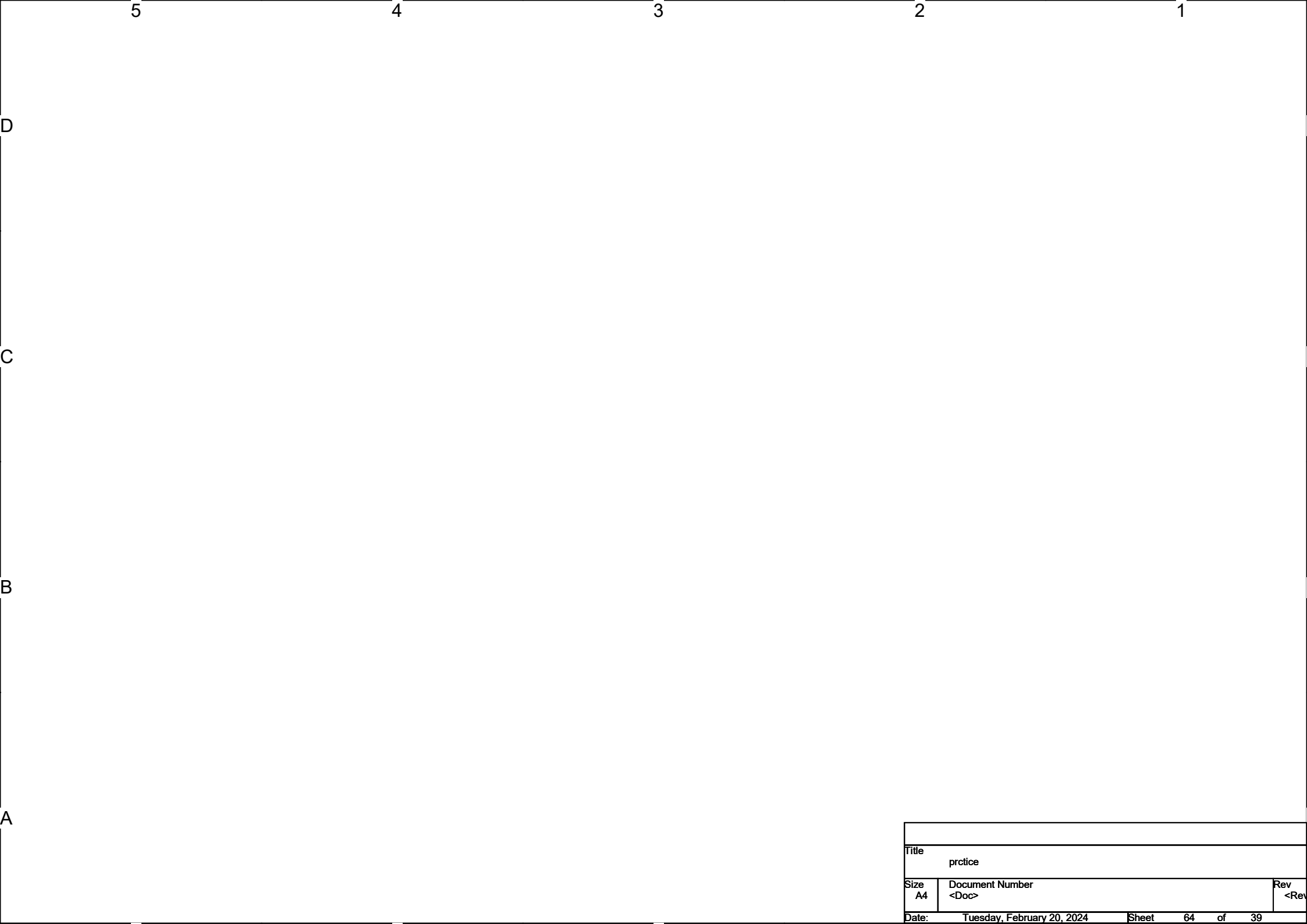


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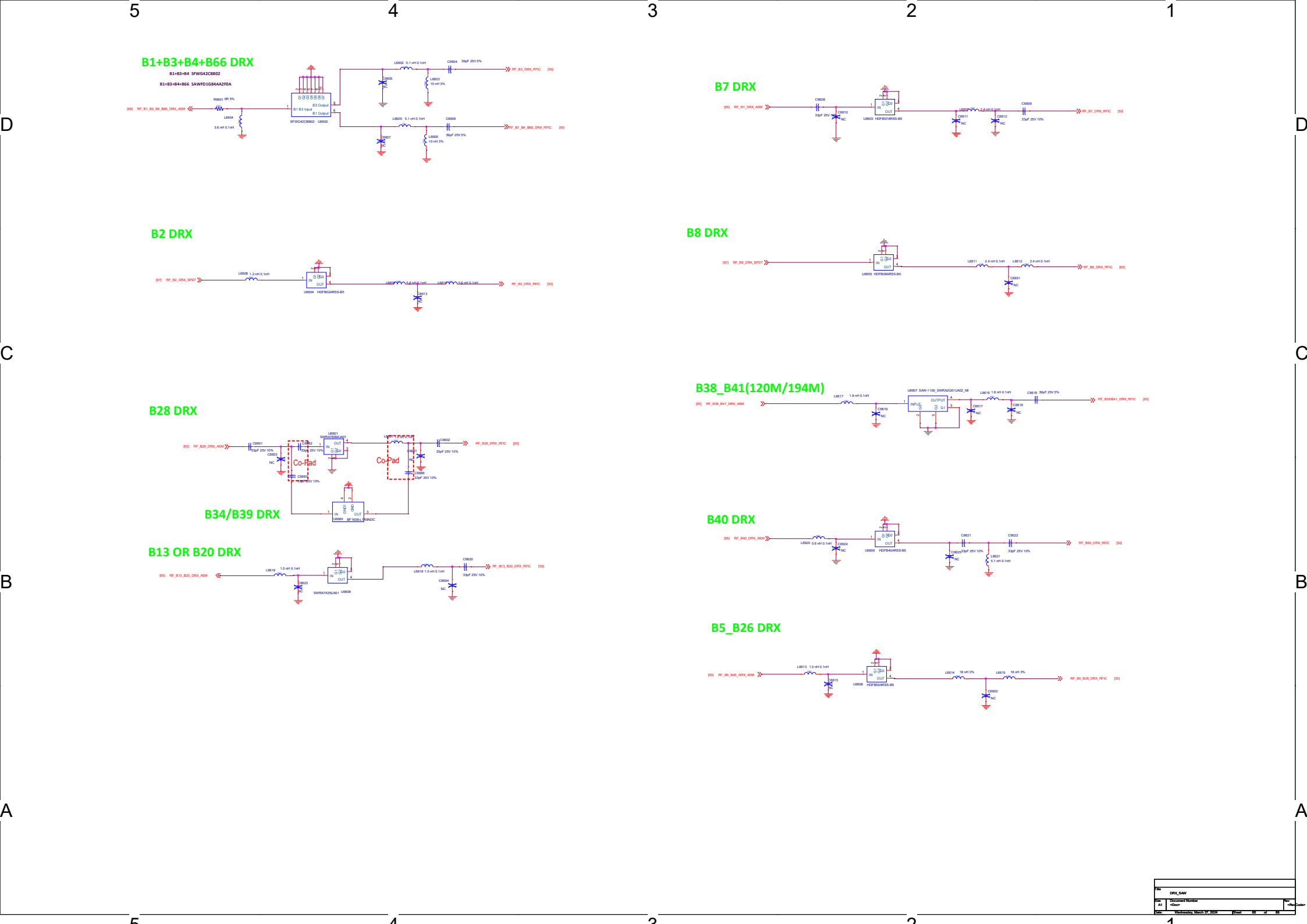


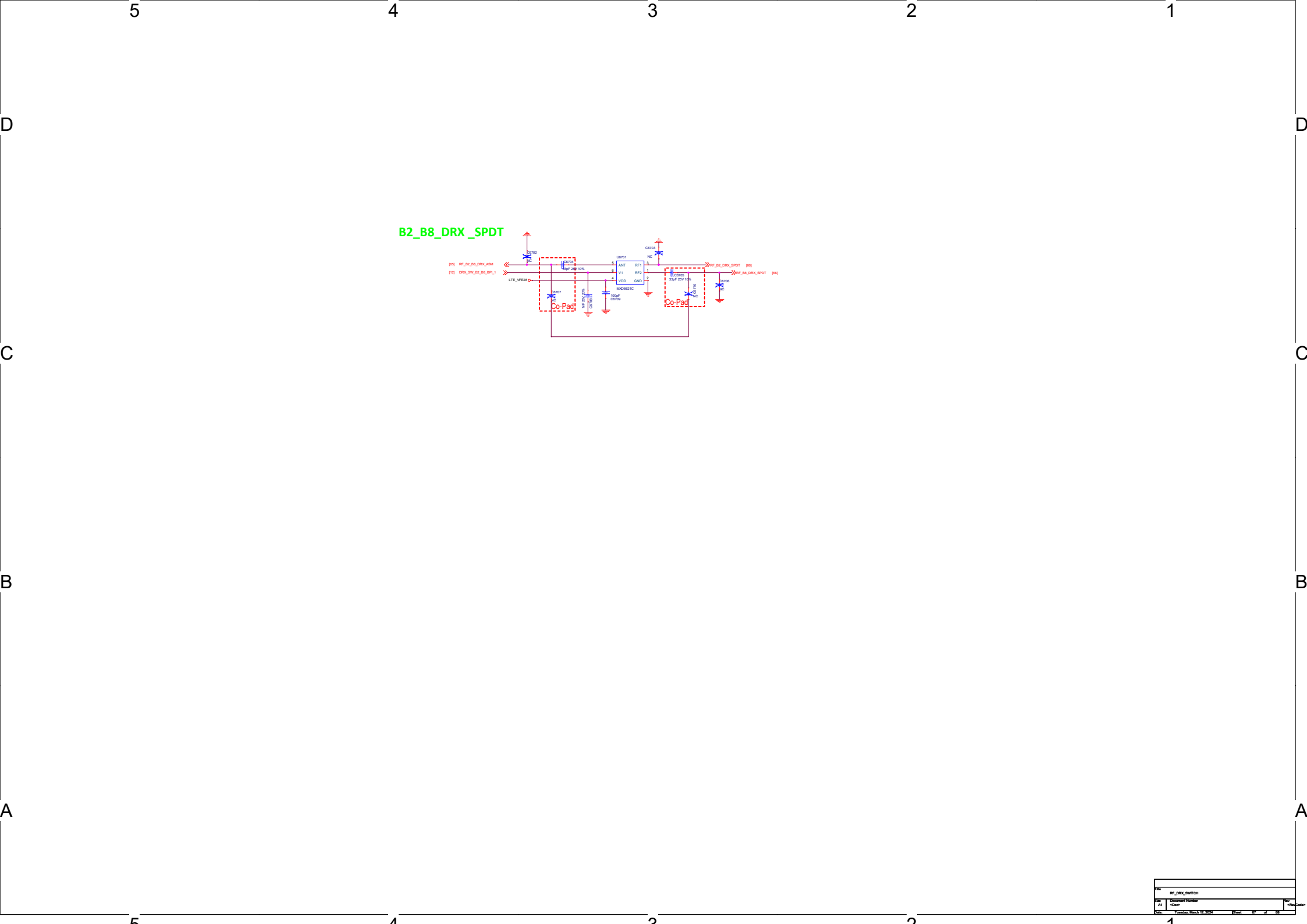
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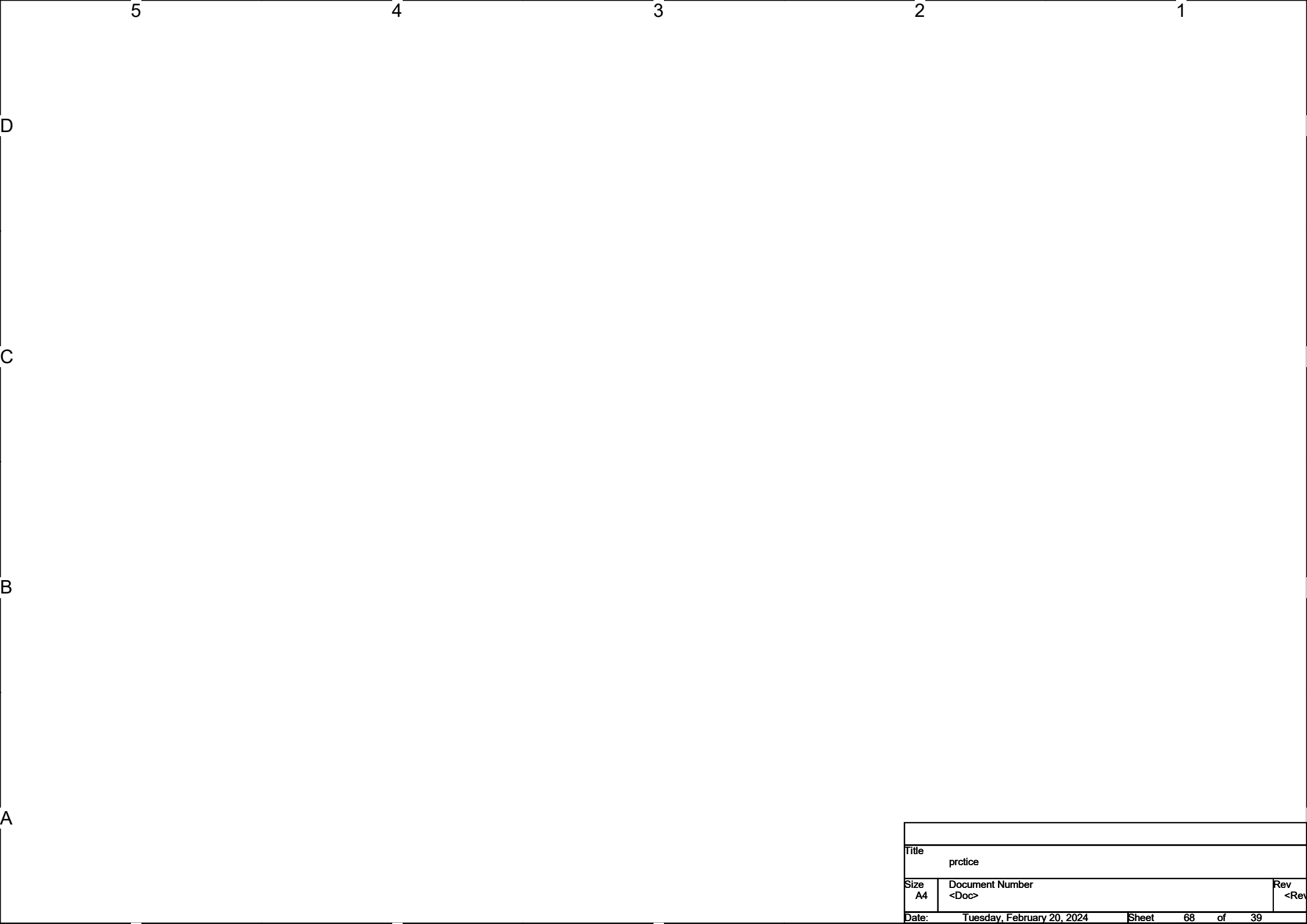
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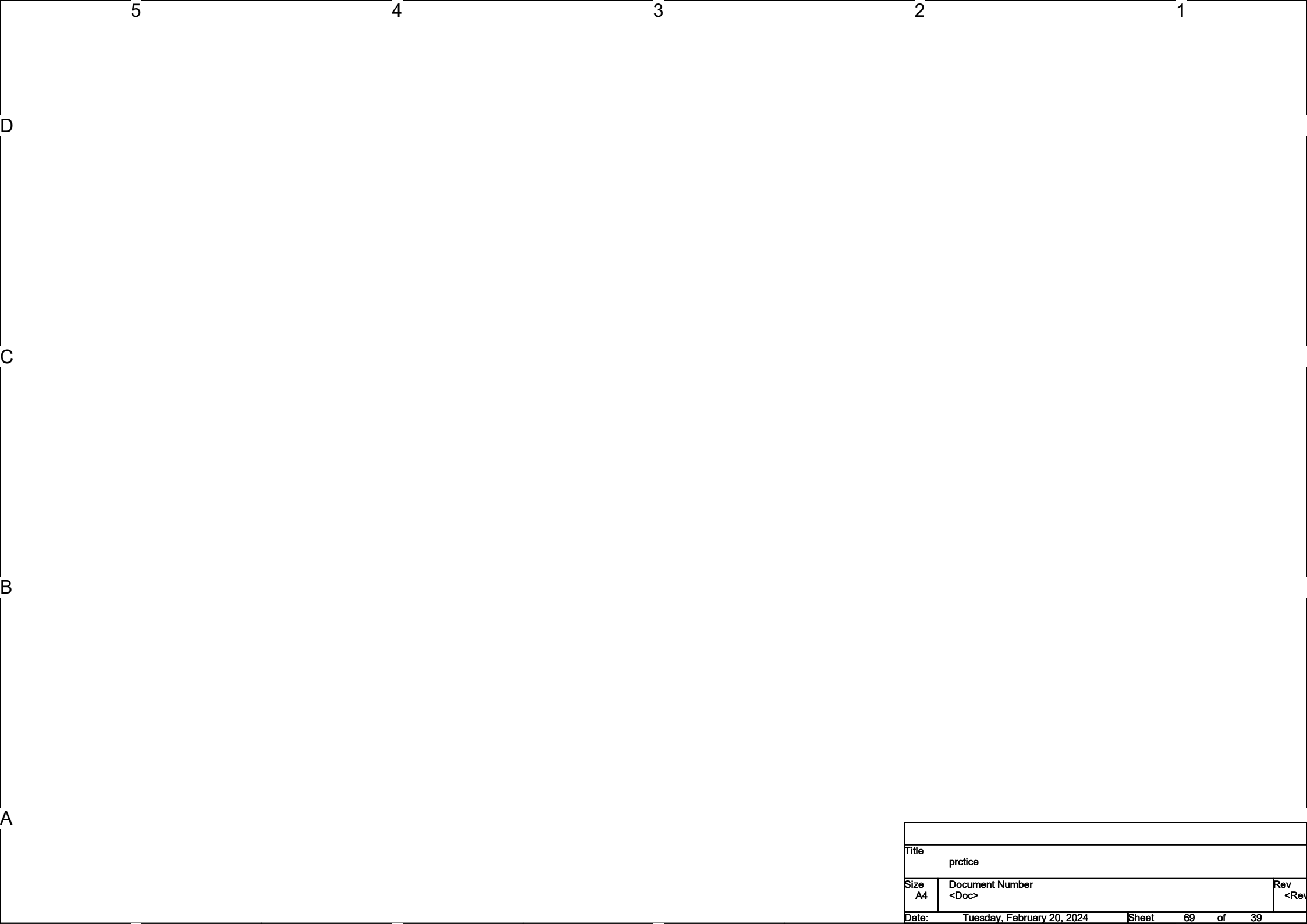
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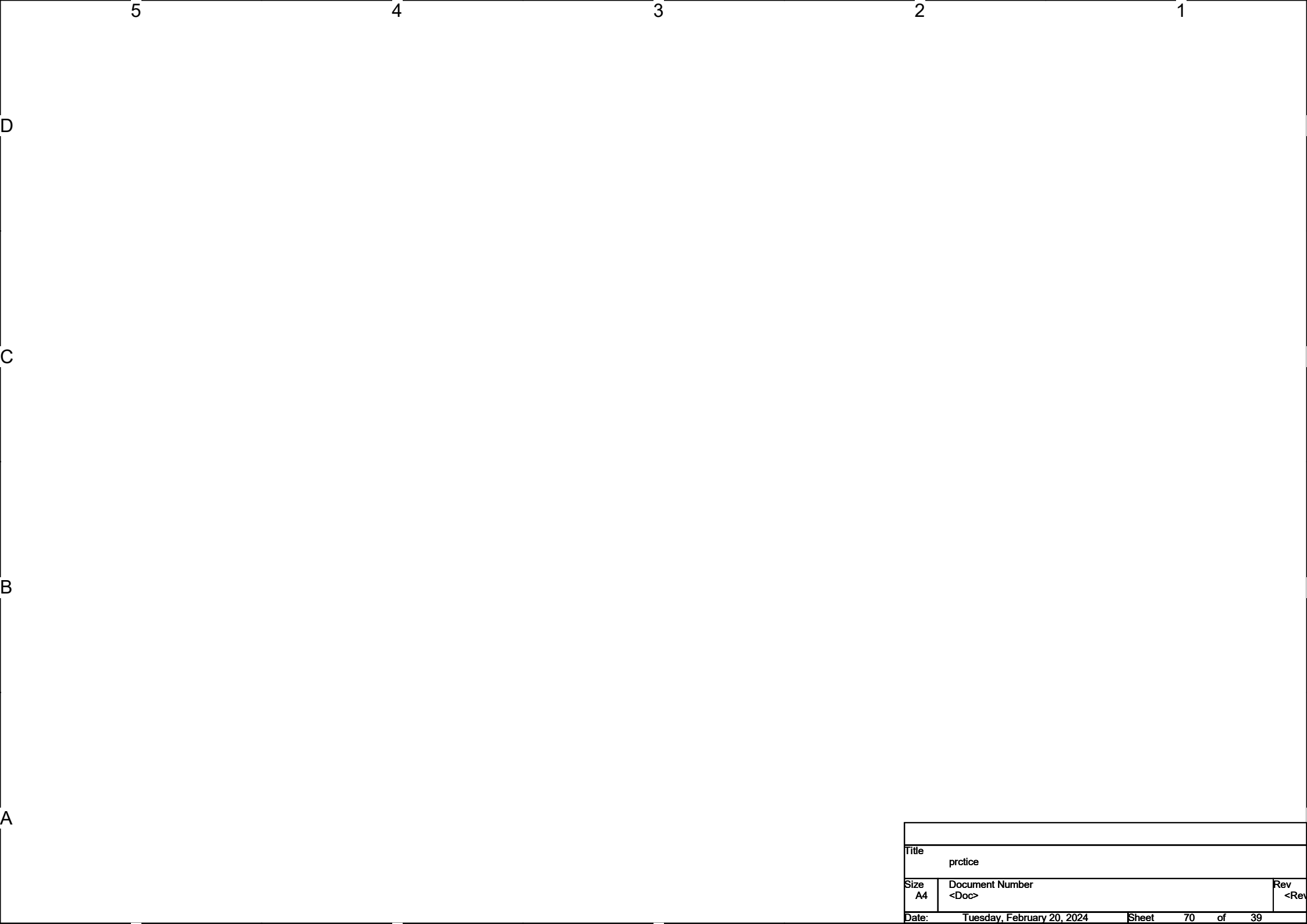




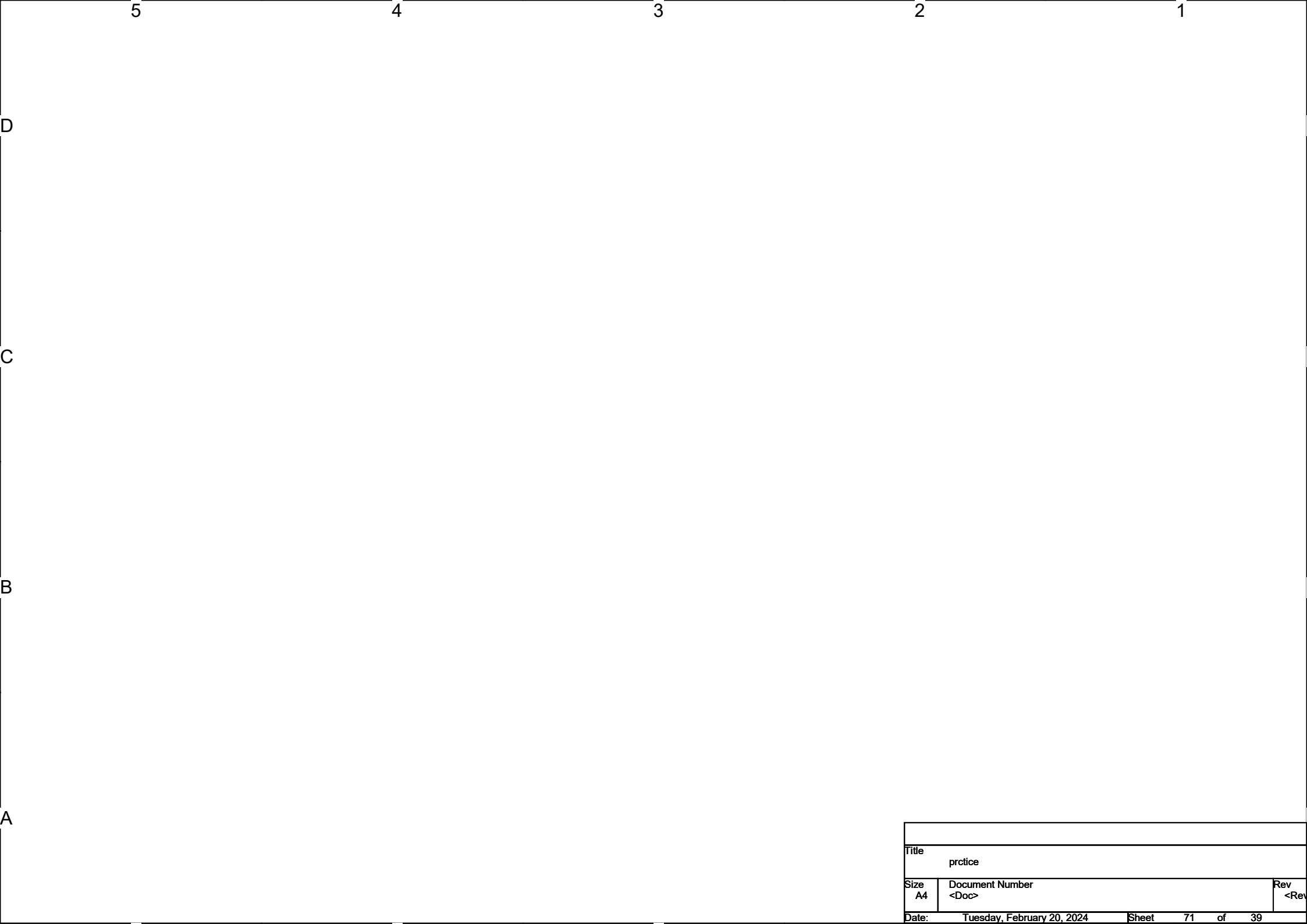
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Date:	Tuesday, February 20, 2024	Sheet 68 of 39



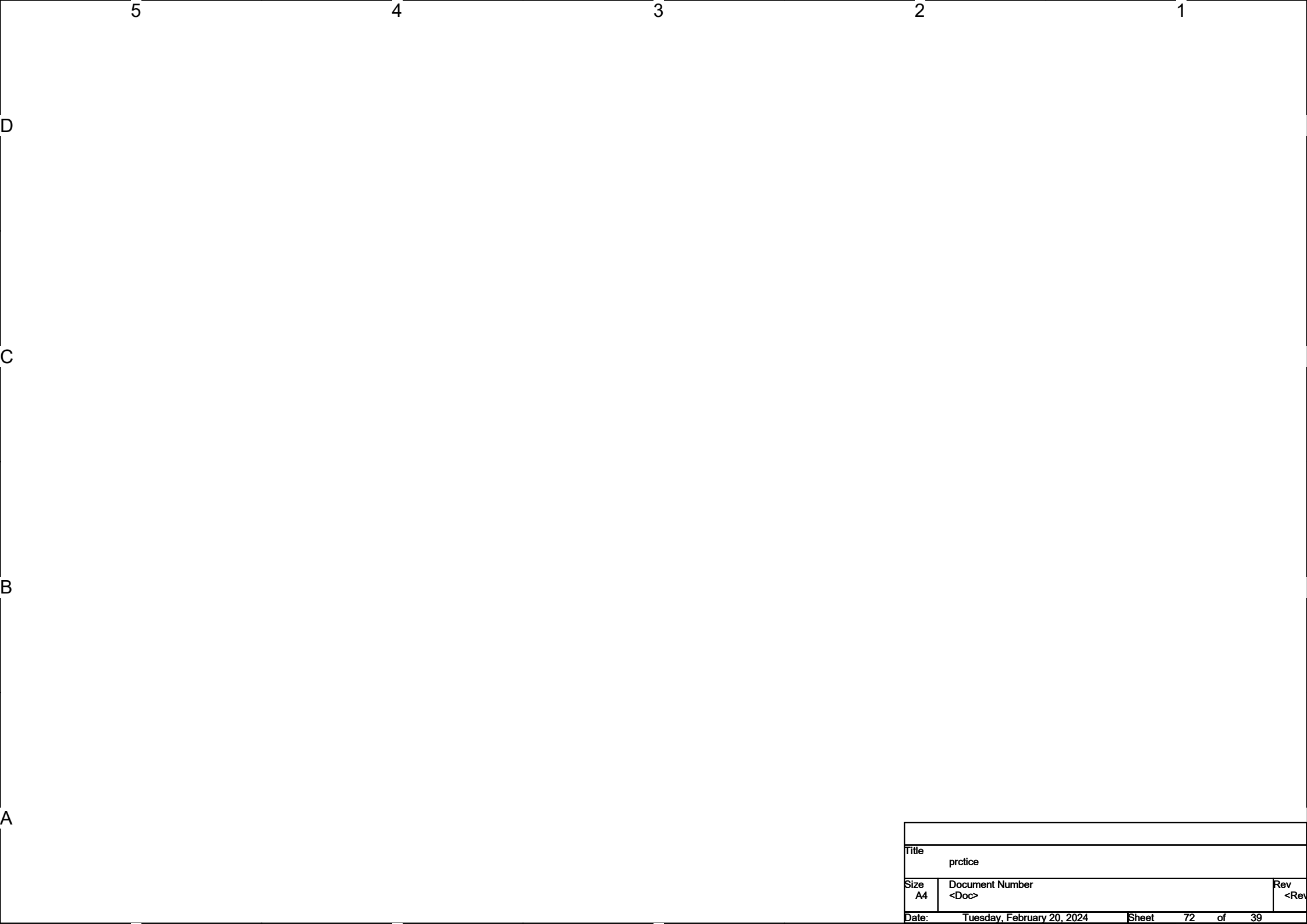
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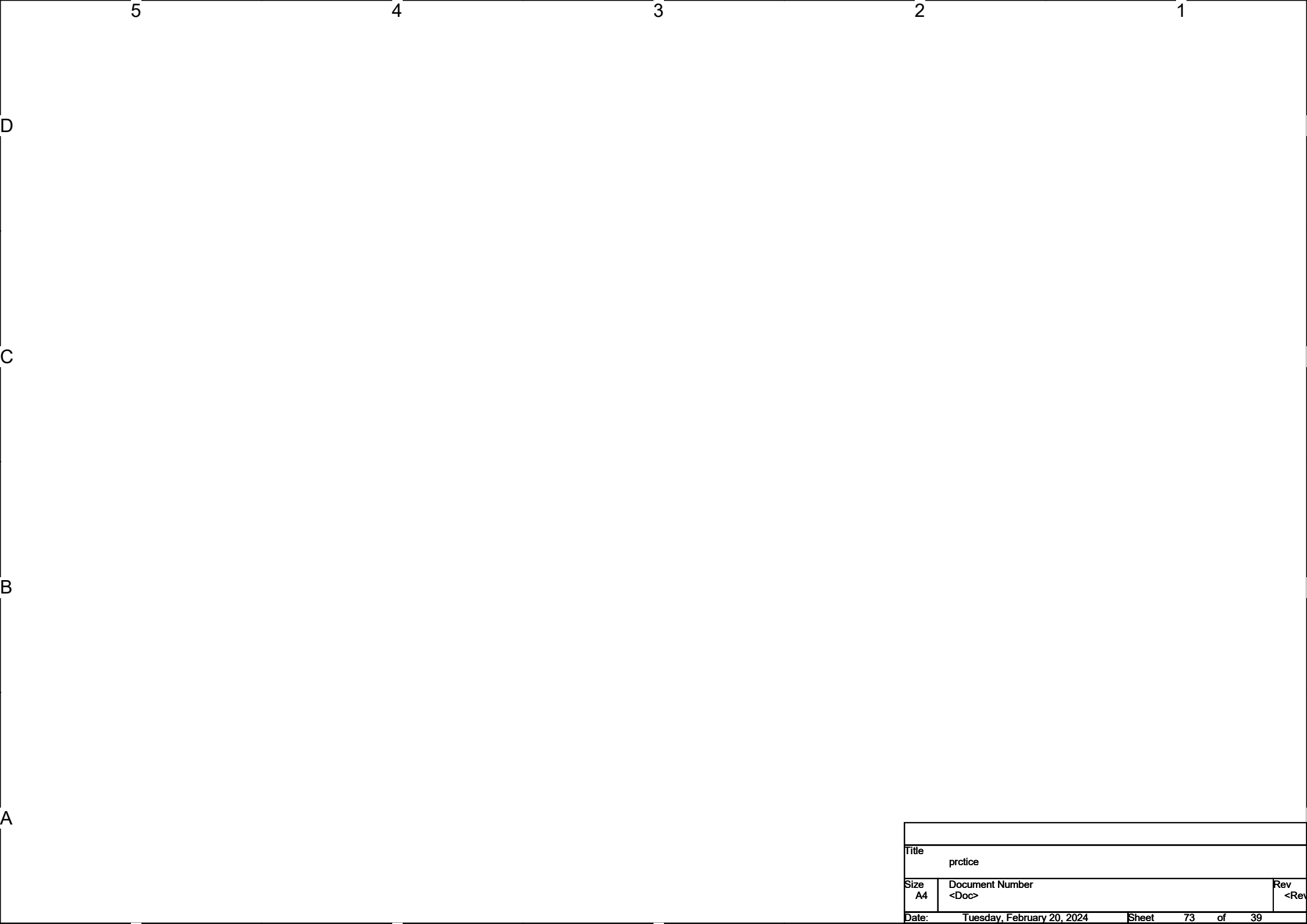
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Date:	Tuesday, February 20, 2024	Sheet 70 of 39



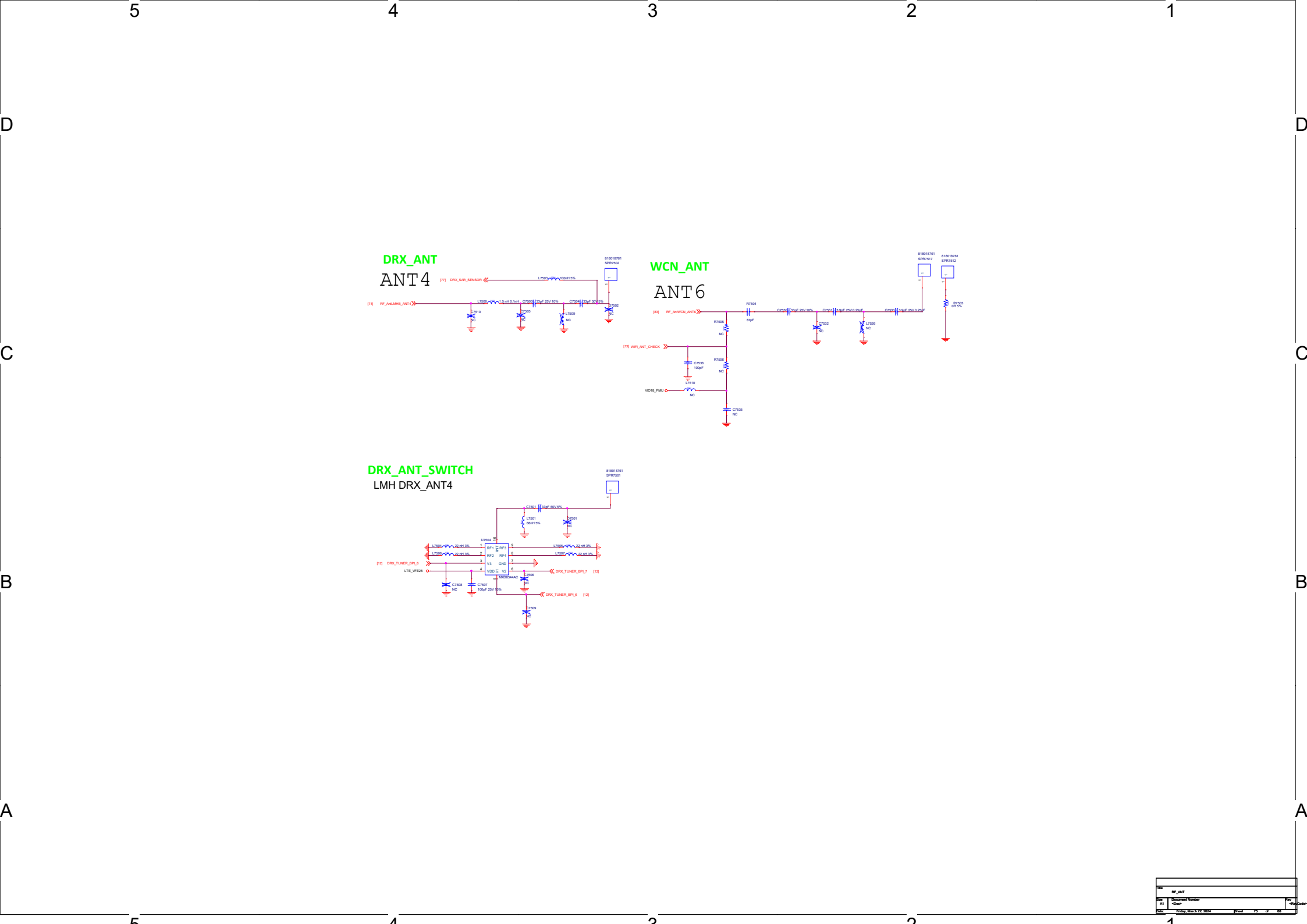
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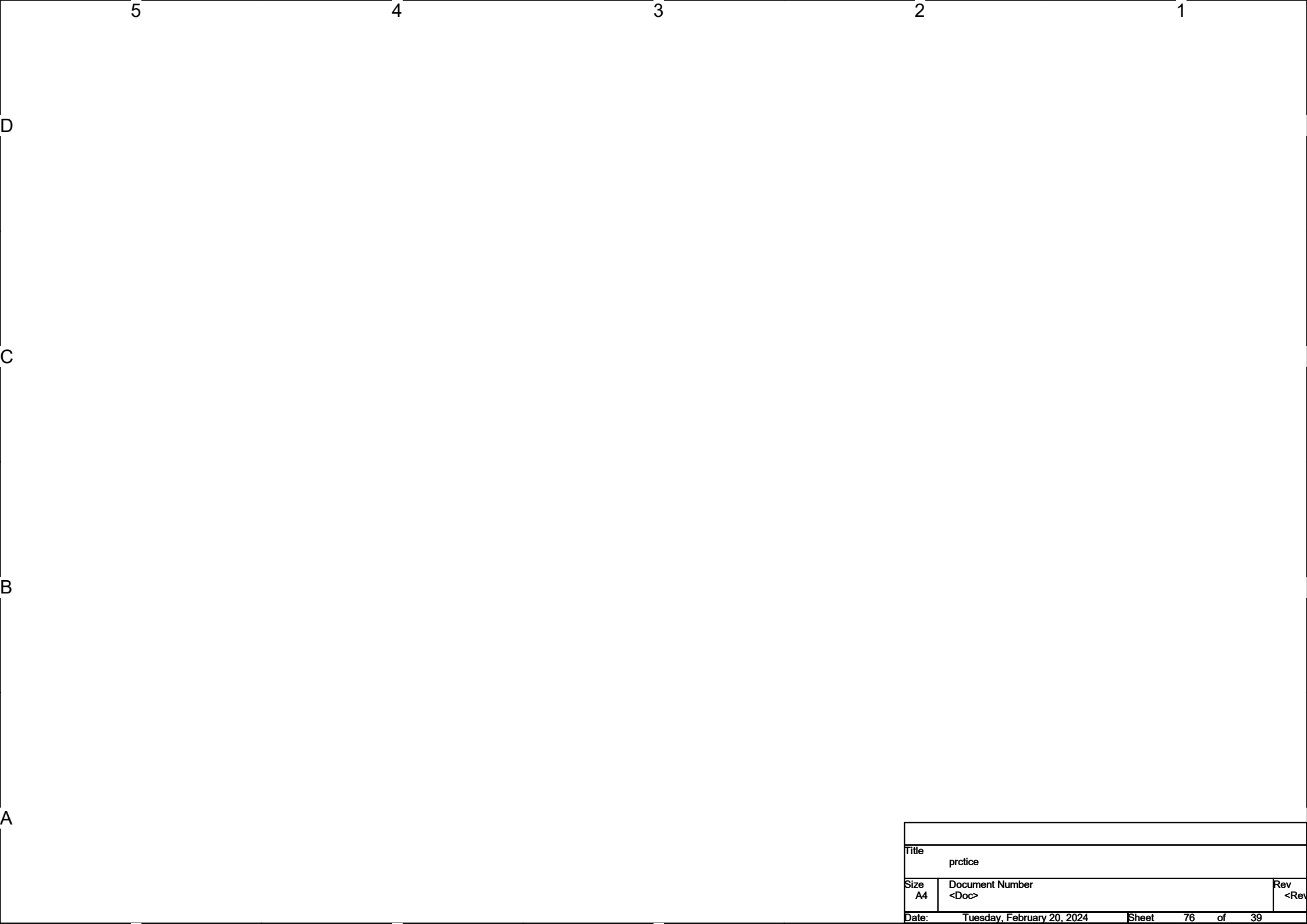


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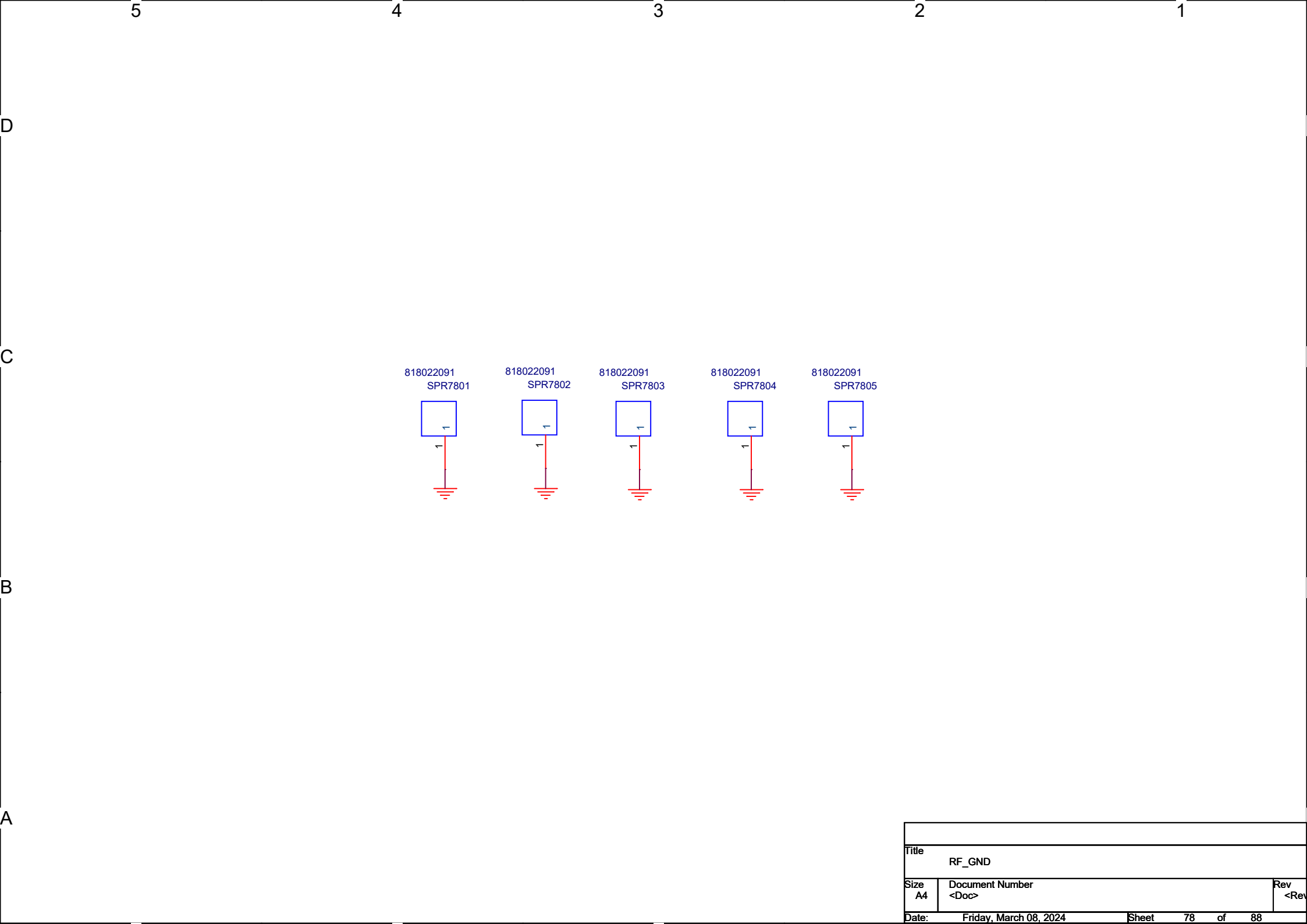
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Date:	Tuesday, February 20, 2024	Sheet 73 of 39



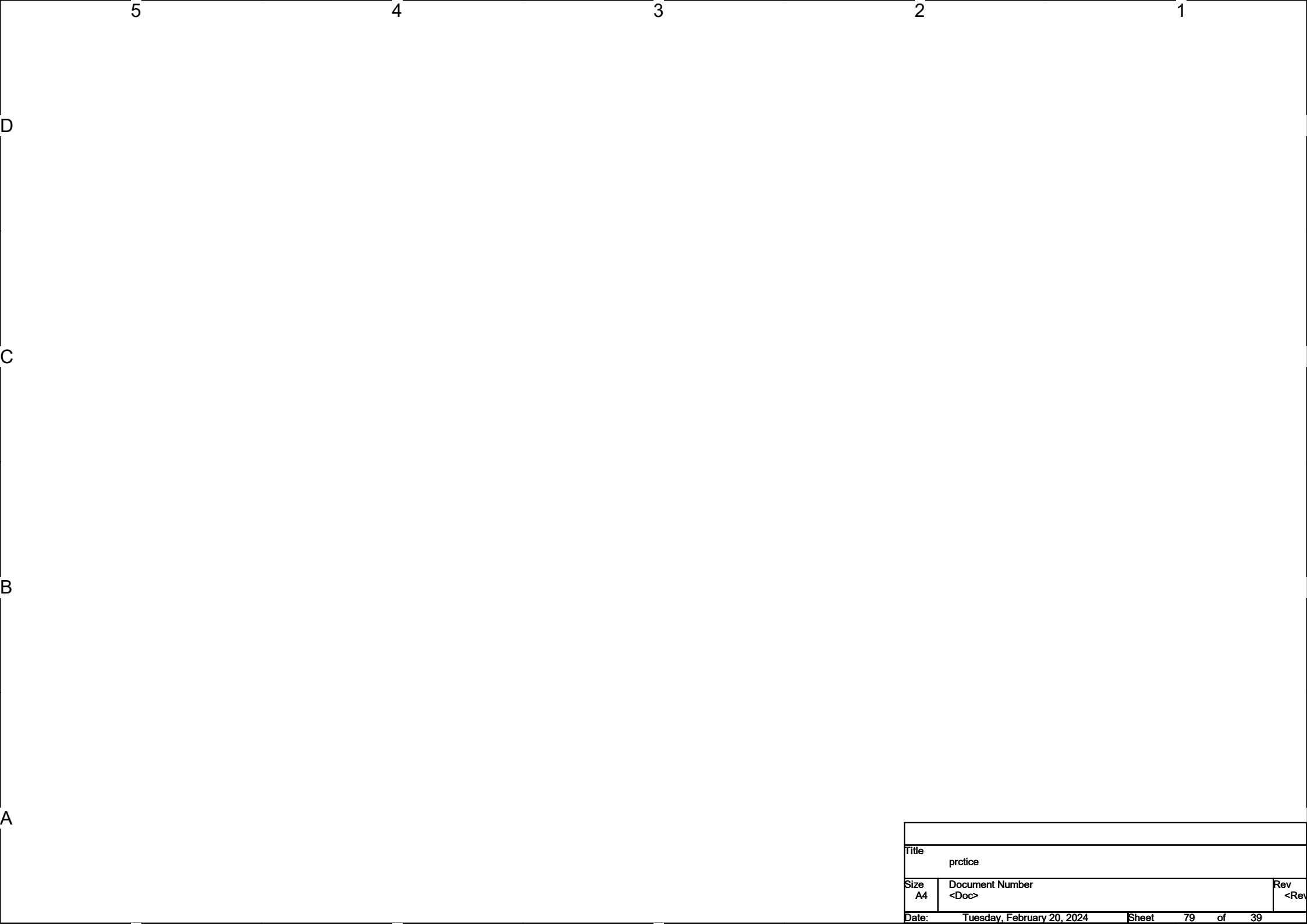


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Size	Document Number	Rev
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Date:	Tuesday, February 20, 2024	Sheet 76 of 39

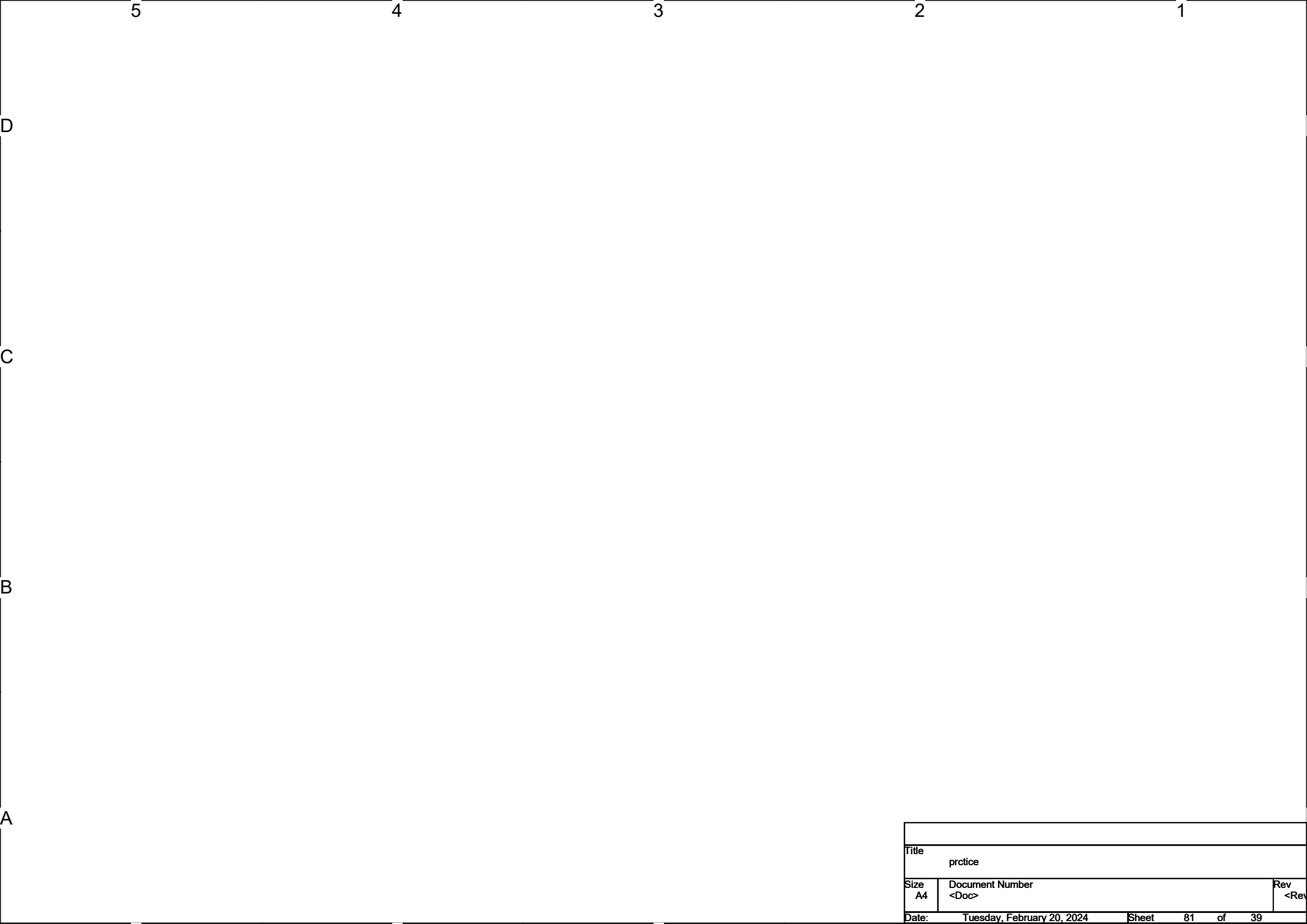




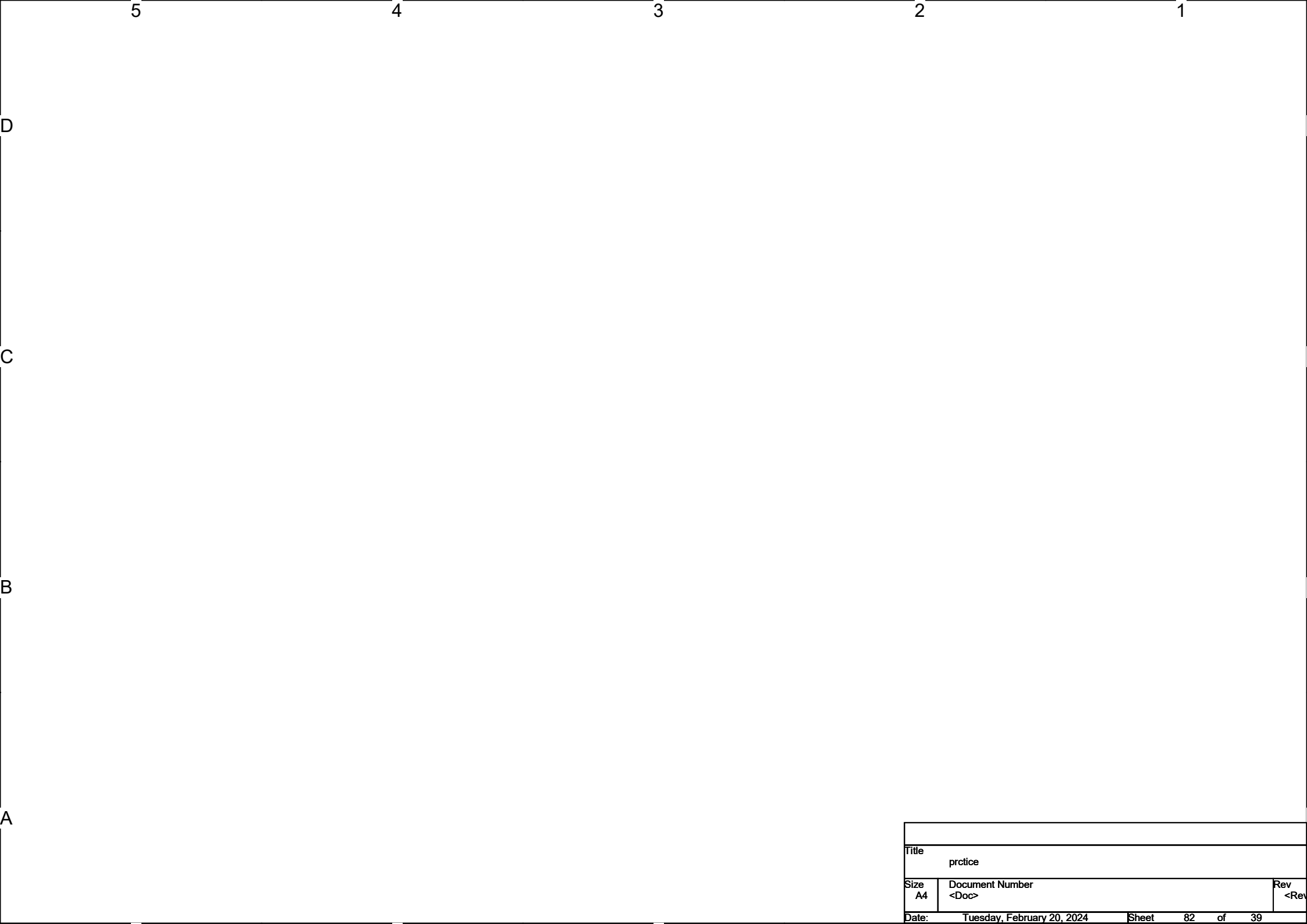
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Size	Document Number	Rev
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Date:	Friday, March 08, 2024	Sheet 78 of 88



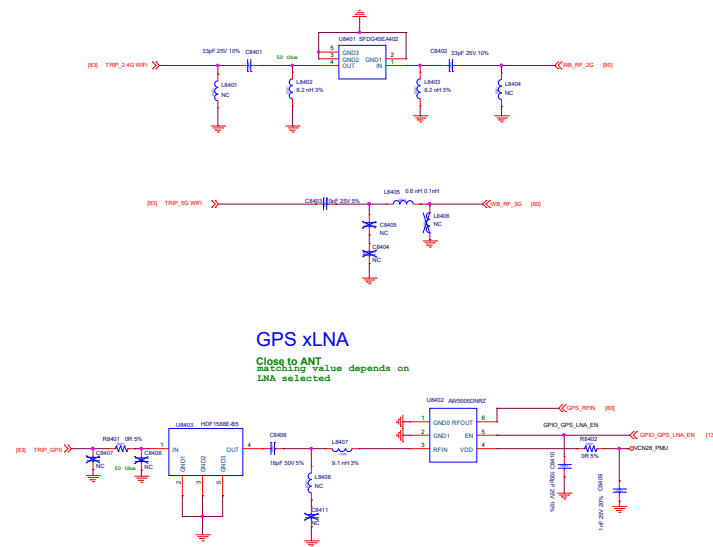
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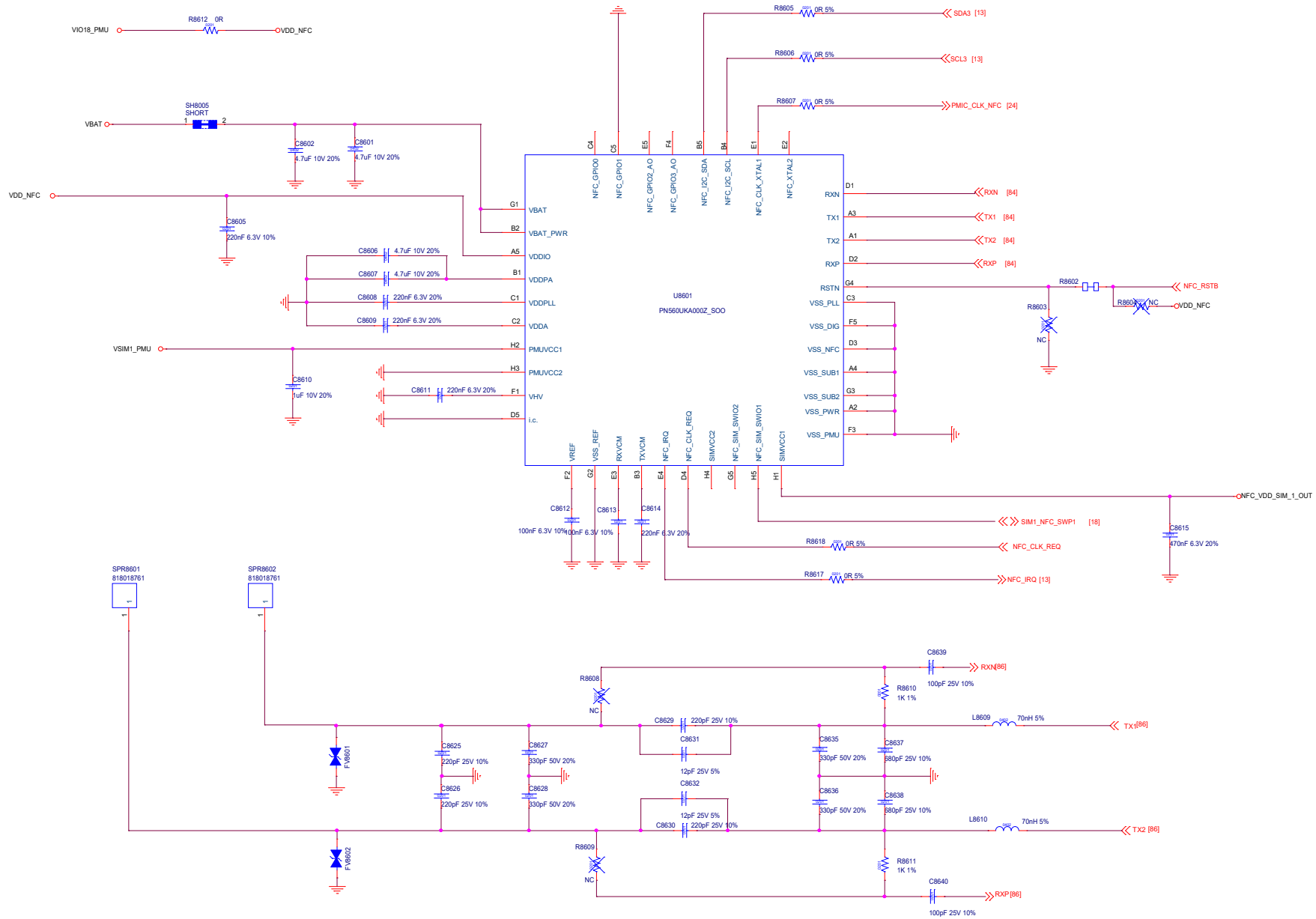


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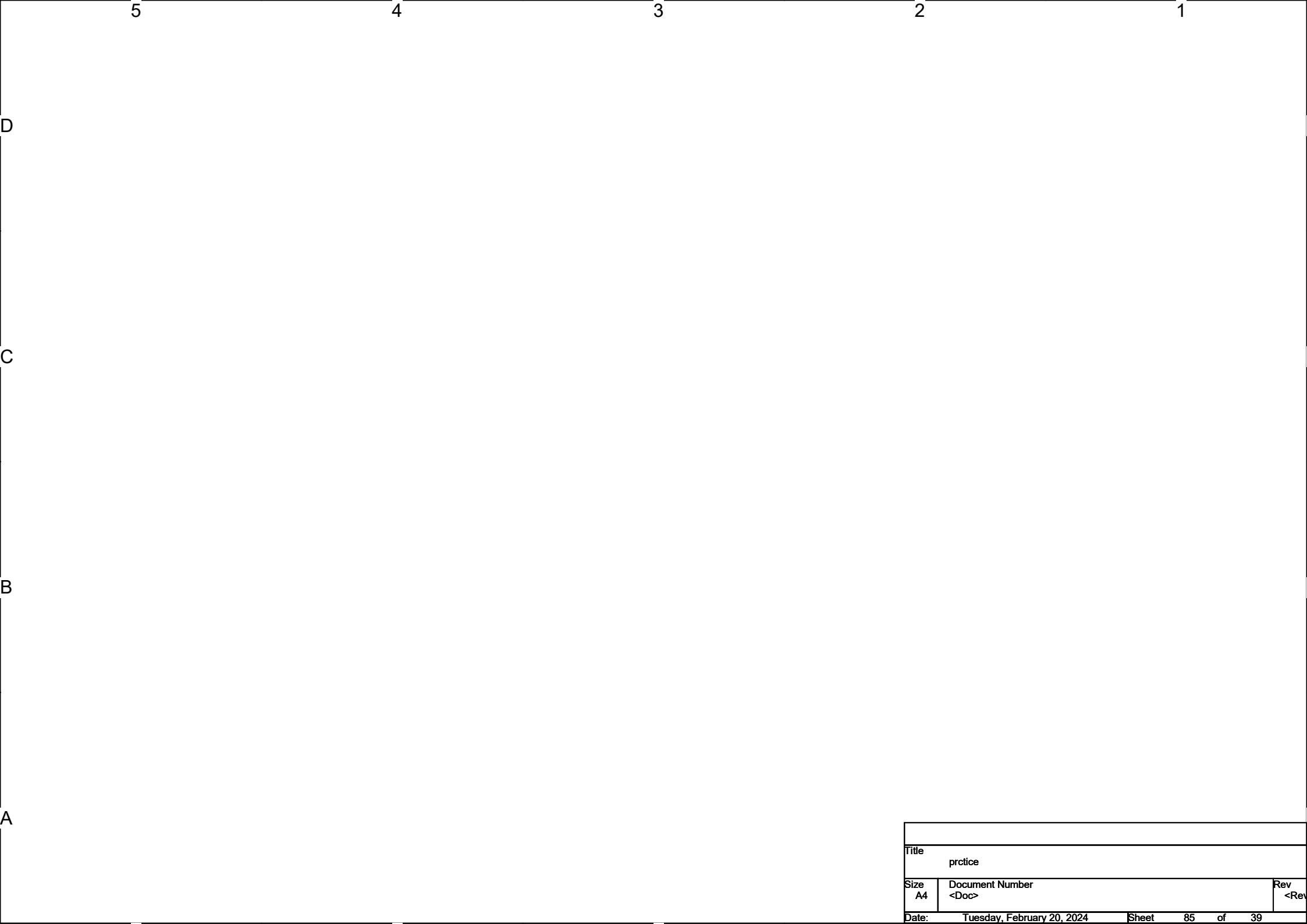


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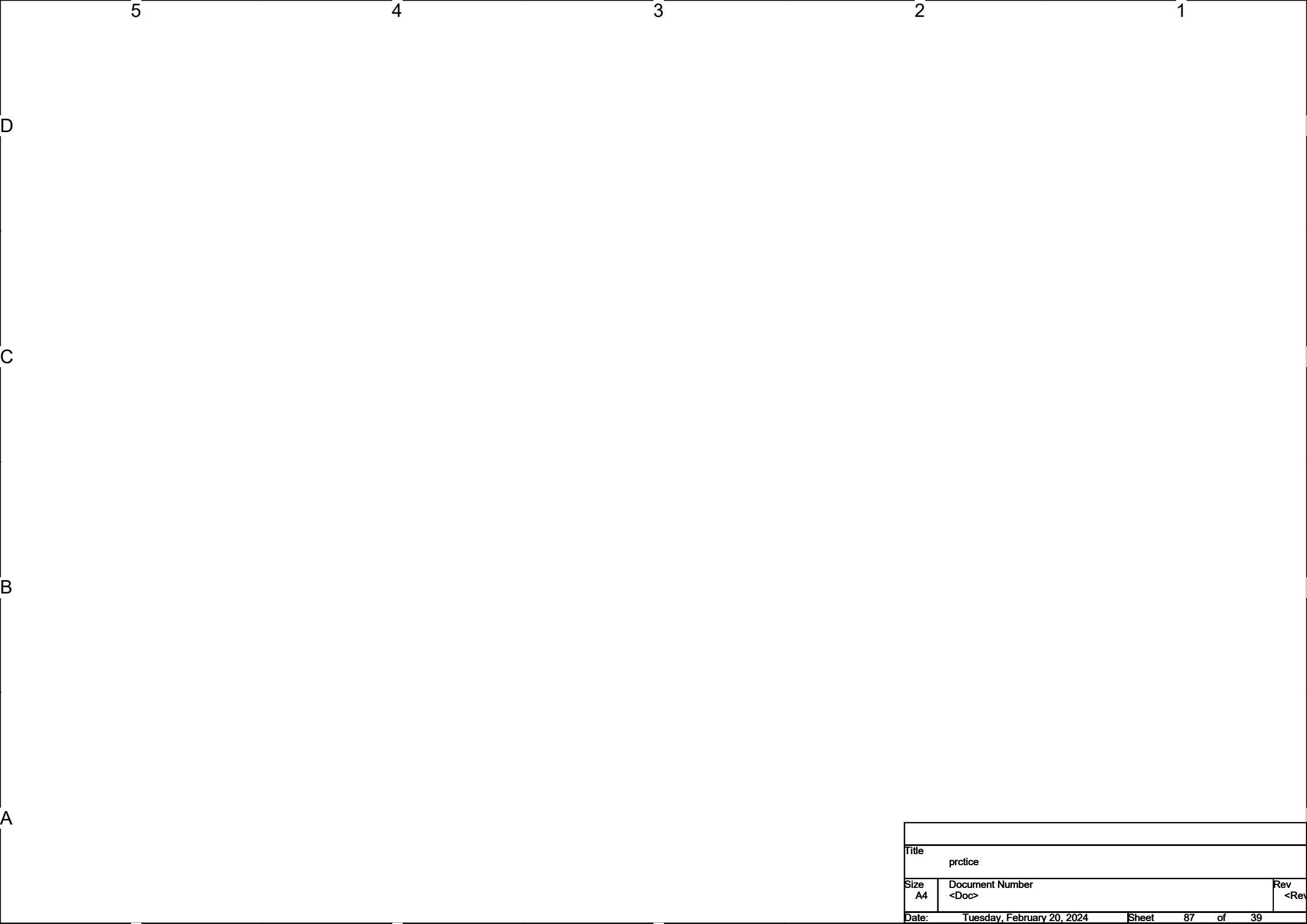




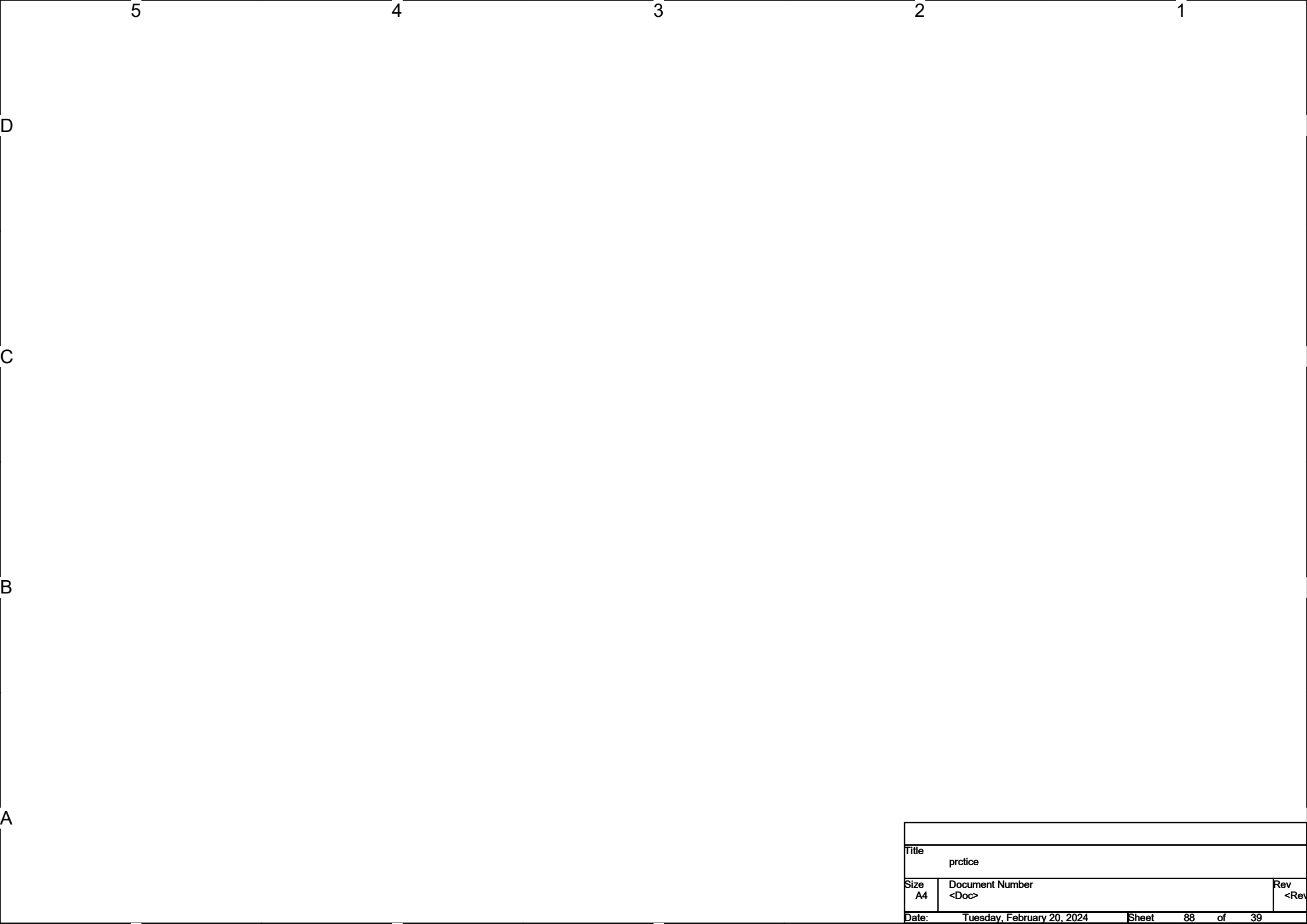
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